

A welfare evaluation of the 1986 tax reform for married couples in the United States

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Abstract This paper evaluates the welfare effects of the 1986 Tax Reform Act (TRA86). We rely on different welfare metrics, which fully retain preference heterogeneity and are based on different ethical priors. We estimate utility functions with preference heterogeneity on the basis of structural models of family labor supply. Then, using these estimated preferences, we compute and compare different well-being rankings corresponding to different ways of measuring well-being. Finally, we identify the losers and the winners of TRA86, in absolute and relative terms, for each of the welfare metrics.

Keywords Welfare measures · Tax reform · Preference heterogeneity · Discrete model · Labor supply

JEL Classification D63 · H31 · J22

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1 Introduction

Since the beginning of the 1980s, the USA went through several reforms of the federal income tax rules. As described in [Eissa et al. \(2008\)](#), a series of tax acts were passed between 1981 and 2003. These reforms defined somehow a trend in the change of the US tax system consisting of (i) the simplification of the tax code; (ii) the decrease in the marginal tax rates; (iii) the increasing importance of the Earned Income Tax Credit (EITC) for low-income taxpayers.

So far, the literature on the evaluation of the US tax reforms has mostly dealt with labor supply effects, especially those due to the increased role of the EITC as a public support for poor families and to the concurrent reduction of welfare benefits. The most recurring question concerns the labor market participation effects induced by the changes in the EITC.¹ Some researchers have also focused on other dimensions. [Eissa et al. \(2008\)](#) evaluated the efficiency effects of the reform.² They found that the 1986, 1990, 1993 and 2001 tax reforms generated relevant efficiency gains, especially through the increased participation of single mothers.

In this paper, we are primarily interested in the welfare effects of TRA86 with a special focus on resource inequality. The recent literature on income inequality has showed that pre-tax income inequality has notably increased over the last 30 years. This conclusion has been first brought forward by [Piketty and Saez \(2003\)](#), who built a series of pre-tax income shares based on tax return data from the Internal Revenue Service (IRS). They specifically highlight the increase of the top 1% of income share. A more recent contribution by [DeBacker et al. \(2013\)](#), also based on IRS data, not only confirms this trend but also stresses its persistent nature. These findings are also confirmed by studies relying on Current Population Survey data ([Heathcote et al. 2010](#); [Burkhauser et al. 2011](#)): income inequality rose sharply in the 1980s and this growth continued at a reduced pace in the 1990s and early 2000. Finally, [Bargain et al. \(2015\)](#) look at the increase in inequality over the last three decades solely due to tax policy changes.³ They find that tax policies alone have an important effect on the evolution of inequality, but they only explain a small part of the significant increase in the income share of the top percentile.

Even if these findings link the evolution of the tax system to the increase in income inequality, they are silent about the distribution of individual well-being. Moreover, most of the literature on applied welfare economics has so far neglected the norma-

¹ As the EITC depends on family income, it is expected to stimulate the labor supply at the extensive margin but, in the phase-out region, it should reduce incentives at the intensive margins. Empirical studies find that the EITC expansions had heterogeneous effects on labor supply, depending on family composition. On the one hand, [Dickert et al. \(1995\)](#), [Eissa et al. \(1996\)](#), [Ellwood \(2000\)](#) and [Meier and Rosenbaum \(2000, 2001\)](#) show that the EITC reforms increased the workforce participation of single women with children, since it made work more attractive. On the other hand, [Ellwood \(2000\)](#) and [Eissa and Hoynes \(2004\)](#) find that the EITC expansion between 1984 and 1996 reduced the total family labor supply of married couples. Overall, the positive effect on labor force participation seems to be large enough to offset the negative impact at the intensive margin in working hours ([Scholz 1996](#)).

² As an efficiency measure they use the excess burden of taxation, i.e., the sum of how much each individual would be willing to pay to get rid of all taxes and transfers.

³ In a similar fashion, [Piketty and Saez \(2007\)](#) analyze changes in the progressivity of the federal income tax over time but cannot disentangle policy changes from other factors.

tive consequences of the reforms or has used conventional measures of individual well-being based on a “reduced” version of the underlying model. For instance, many applied papers report only aggregate labor supply changes or, when it comes to distributional issues, they use disposable income as a proxy of individual well-being. Other papers rely on more sophisticated measures of well-being, such as the equivalent income, but they just consider the underlying preferences of a reference household so that heterogeneity in preferences is assumed away (King 1983; Aaberge et al. 2004; Aaberge and Colombino 2013).

We intend to bridge this gap by introducing individual heterogeneous preferences at the core of our analysis. We rely on well-being indexes that fully account for different individual consumption–leisure preferences. Such indexes have been studied in detail by Fleurbaey and Maniquet (2011). They propose an axiomatic construction so that one can clearly see the normative considerations and the ethical priors behind the ways interpersonal comparisons are performed, while still retaining preference heterogeneity. Different ethical priors, concerning the weight to be given to individual circumstances and to individual preferences, yield different ways of measuring individual well-being. We intend to use these notions of individual well-being in order to evaluate the effects of TRA86 under different ethical priors.

Our contribution is threefold. First, to the best of our knowledge, this is the first attempt to evaluate a full-fledged fiscal reform using different ideas of fairness embedded in the particular measure of individual well-being that is employed.⁴ Second, in order to gain a deeper insight into the welfare impact of the reform, we decompose the utility gains into a direct income component, due to the change in the tax schedule, and a behavioral component, due to individuals’ labor supply responses. Third, when estimating heterogeneous households’ preferences for consumption and leisure, we take advantage of panel data to control for time-constant unobserved heterogeneity in a discrete choice structural labor supply model. This allows us to estimate more accurately the marginal rates of substitution between income and leisure and the importance of some observed characteristics in determining their difference.

In our empirical analysis, we use the 1984–1991 Panel Study of Income Dynamics (PSID) and focus only on married couples. We find that TRA86 increased the well-being of 99.7% of the families in our sample. This is essentially due to the fact that TRA86 enlarged the budget constraint for virtually everybody (however, there is still a small fraction of households, 0.9%, who faced an increase in well-being jointly with a reduction in their net income). We hence specifically focus on relative gains: we find that TRA has determined a significant redistribution of income and well-being between the lower-middle class and the upper-middle class. However, if one specifically looks at different welfare measures, the intensity of such redistribution seems strongly related to the weight that the policy maker gives to individual (heterogeneous) preferences.

This is even more evident when we decompose the utility gains into an income component and into a behavioral component. We find that the households at the bottom

⁴ Bargain et al. (2013), Carpentier and Sapata (2016) and Decoster and Haan (2015) also use well-being indexes that are similar to those which we employ here. They do it, however, with different purposes: while Bargain et al. (2013) focus on a cross-country comparison of well-being, the other two studies deal with the sensitivity of different well-being rankings (in the USA and Germany) to different normative principles.

of the well-being distribution lost positions in the social ranking mainly because of their behavioral response, while they mostly gained because of the increased income granted by the reform. The behavioral component and the income component tend to compensate each other, and their magnitude varies greatly depending on the ethical priors embedded in the chosen well-being index. The level of well-being and its variation are indeed differently responsive to the shape of the entire indifference maps depending on the welfare metric we use.

Our study is organized as follows. Section 2 describes the tax-benefit system in the USA before and after TRA86. Section 3 presents the well-being indexes used to evaluate the impact of TRA86 on welfare. Section 4 deals with the estimation of household preferences. Section 5 presents the data, reports the estimation results of household preferences and studies the impact of TRA86 on households' well-being. Section 6 concludes.

2 The 1986 tax reform act

One of TRA86's main features was the simplification of the tax code through the drastic reduction in the number of the marginal tax rates' brackets. Their number was reduced from 15 to 5 in 1987 and then to 4 from 1988 until 1990. Table 1 reports the main features of the federal income tax structure from 1984 to 1990 (for married couples filing jointly). TRA86 increased the lower marginal tax rate from 0 to 15% but compensated the removal of the zero tax rate bracket by increasing the standard deductions and, above all, the personal exemptions, which doubled in nominal terms between 1984 and 1990. Another important feature of TRA86 is the decrease in the highest marginal tax rate. Finally, TRA86 made the EITC more generous. It increased the phase-in rate, decreased the phase-out rate and enlarged the maximum tax credit. Indeed, between 1984 and 1990 the maximum credit and the maximum earnings still entitling taxpayers to the tax credit almost doubled (in nominal terms).

Figure 1 summarizes, on the left panel, the relation between real earnings and federal income taxes, for married couples, before and after TRA86 for household earnings spanning from \$0 up to \$22,000. Similarly, the right panel summarizes the relation between real earnings and total taxes, federal income, state income and Federal Insurance Contributions Act (FICA) taxes, in California (the most populated state in the USA). It is clear that TRA86 reduced the average tax rate of both couples with kids and without kids.⁵ However, not all the types of households are impacted with the same intensity: lower earnings couples did not see their situation changing very much, especially if without kids. The biggest change in taxation involves gross earnings of about \$10,000. If we focus on federal income taxation, married couples enjoy the largest improvement, essentially induced by the larger generosity of the EITC. Couples without kids also benefited, to a lesser extent, from a reduction in the average tax rate.

⁵ If we run the same simulation for earnings larger than \$200,000, we arrive at the same conclusion. The only difference is that the reduction in the taxes at federal and state level induced by TRA86 is increasing with earnings in absolute value.

Table 1 Federal income tax parameters from 1984 to 1990 for married couples filing jointly (nominal dollars). *Source:* Tax Foundation and authors' calculations from the Tax Simulation Model (TAXSIM) of the National Bureau of Economic Research (NBER), available online at <http://www.nber.org/taxsim/>

Tax year	Marginal tax rates (%)		Number of brackets	Bottom bracket (\$)	Top bracket (\$)	Standard deduction married couples (\$)	Personal exemptions for the couple (\$)
	Minimum	Maximum					
1984	0.0	50.0	15	0–3400	> 162,400	3400	2000
1985	0.0	50.0	15	0–3540	> 169,020	3550	2080
1986	0.0	50.0	15	0–3670	> 175,250	3670	2160
TRA86							
1987	11.0	38.5	5	0–3000	> 90,000	3760	3800
1988	15.0	33.0 ^a	4	0–29,750	> 149,250	5000	3900
1989	15.0	33.0 ^b	4	0–30,950	> 155,320	5200	4000
1990	15.0	33.0 ^c	4	0–32,450	> 162,770	5450	4100

^a A 33% “rate bubble” applied between \$71,900 and \$149,250, the applicable tax rate was otherwise 28% for earnings above \$29,750

^b A 33% “rate bubble” applied between \$74,850 and \$155,320 the applicable tax rate was otherwise 28% for earnings above \$30,950

^c A 33% “rate bubble” applied between \$78,400 and \$162,770 the applicable tax rate was otherwise 28% for earnings above \$32,450

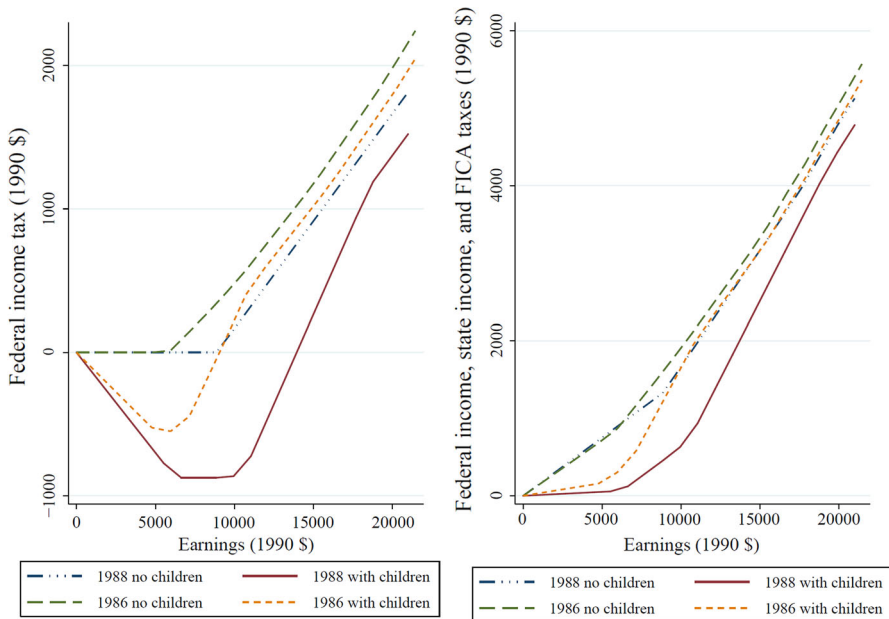


Fig. 1 Tax schedules before and after TRA86 by the presence of kids under age 17. *Notes:* the simulation of the tax schedule is for married couples filing jointly the tax declaration in California with no taxpayers over 65 years of age, no dividend and other property income, no pensions, no social security benefits, no other non-taxable transfer income, no rent, no real estate taxes, no child care expenses, no unemployment compensation, no other itemized deductions and zero short- and long-term capital gains. *Source:* authors' calculations from TAXSIM

3 Well-being indexes and their normative framework

The study of the effect of a tax reform on the well-being of individuals with heterogeneous preferences and skills (hence different wage rates) is not easy. Usually, in the public economics literature, it implies the use of weighted utilitarian social welfare functions (Fleurbaey 2008). However, this typically relies on the arbitrary cardinalization of the agents' utility functions (Stiglitz 1987), while the weights cannot be transparently connected to explicit ethical values (Fleurbaey and Maniquet 2015).

Individuals choose bundles of net income y and leisure l , normalized between 0 and 1, from their budget constraint modified by the tax function, so as to maximize their utility. If we observe individual i , with $i = 1, \dots, N$, choosing the bundle (y_i, l_i) , then

$$(y_i, l_i) = \arg \max \{U_i(y, l) \mid y \leq w_i(1 - l) - T[w_i(1 - l); \mathbf{z}_i], 0 \leq l \leq 1, y \geq 0\}, \quad (1)$$

where U_i is the utility function representing the preferences over leisure and consumption and $T(\cdot)$ is the tax function, which transforms income into taxes, given the wage rate w_i and other individual characteristics $\mathbf{z}_i$. Individuals differ from each other because of: (i) different preferences; (ii) different personal productivity, measured by the wage rate w_i ; (iii) different initial endowments of personal characteristics $\mathbf{z}_i$.

TRA86 modified the tax function $T(\cdot)$ and hence the budget set of individuals. Our aim is to evaluate how the well-being of individuals was affected, both in absolute and relative terms. In order to do so, we need to define a way, or different possible ways, to measure individual well-being. We wish to use indexes that fully respect individual preferences and are based on well-defined ethical priors, so that the interpersonal comparison of individual situations is not the result of an arbitrary exercise.⁶ A compromise solution often used in the literature consists of plugging the chosen bundles into a reference preference ordering, which is then eventually used to compute utilities. Individual preferences still play a role in the determination of the chosen bundles but, in practice, preference heterogeneity does not play any role in the normative phase of the analysis, as the choice of a reference preference imposes a specific cardinalization of utility functions. This implies that the way in which interpersonal comparisons of utilities are performed is not founded on an explicit ethical theory. Indeed, by keeping preference heterogeneity in the normative part of the analysis, one needs to deal explicitly with the fact that individual outcomes are the result of both individual choices, and hence preferences, and individual characteristics, most notably individual productivity. For example, when comparing the situation of two individuals producing the same amount of earnings, one needs to take into account both the fact that they have different preferences, so they may have decided to work a different amount of time, and a different (innate) ability. Which of the two individuals will be considered to be better off depends on the weight the ethical observer attaches to each of these factors.

One of the objectives of the so-called fair income tax approach (see [Fleurbaey and Maniquet 2015](#) for a survey) has been the construction of utility functions and their aggregation embedding ethical principles supporting the idea that various sources of income inequalities do not equally call for redistribution. Personal outcomes not only depend on innate circumstances, but also on individual choices. Individuals should be considered responsible for the latter, but not for the former. This entails that inequalities solely due to innate circumstances should be removed. This is the so-called *principle of compensation*. But also, inequalities due only to individual preferences are legitimate, as individuals should be held responsible for their goals. This is the so-called principle of *neutrality/responsibility*. Both principles are logically independent and to some extent in conflict with each other (see, among others, [Fleurbaey 2008](#); [Fleurbaey and Maniquet 2011, 2015](#)).

We focus on indexes that rely on the principle of compensation to make interpersonal comparisons of individual situations. In practice, we use a family of money-metric utility functions that, in our framework, can be defined as the value of the expenditure function, for a reference wage rate $\tilde{w}$, and the utility level $U_i(y_i, l_i)$:

$$m_i(\tilde{w}; y_i, l_i) = \min\{t \in \mathbb{R} | y = t + \tilde{w}(1-l), U_i(y, l) \geq U_i(y_i, l_i), 0 \leq l \leq 1, y \geq 0\}, \quad (2)$$

⁶ A deeper analysis of the indexes presented here is in [Fleurbaey \(2008\)](#), [Fleurbaey and Maniquet \(2011\)](#) and [Fleurbaey and Blanchet \(2013\)](#).

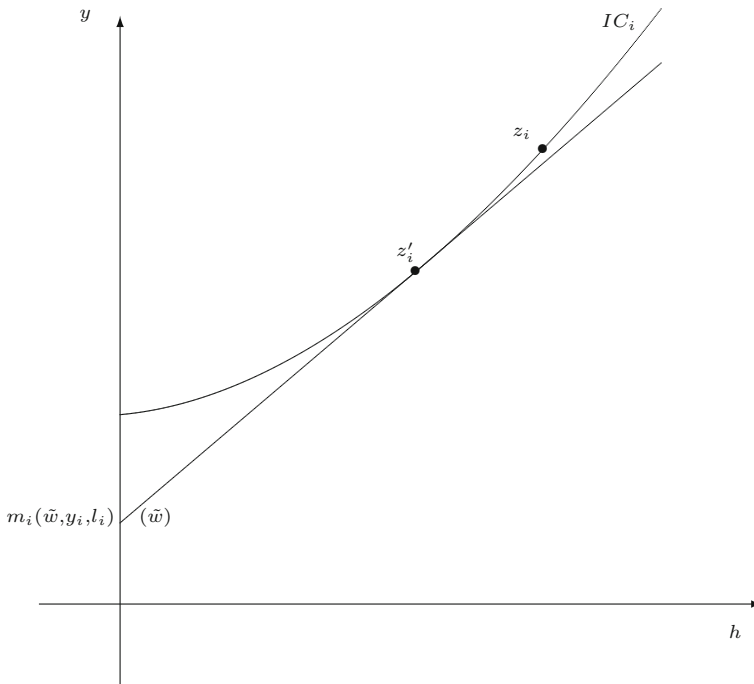


Fig. 2 How to compute the well-being level associated with the bundle z_i for a reference wage rate $\tilde{w}$ and preferences R_i

where t denotes a lump sum transfer. Intuitively, this is the amount of money that makes individual i indifferent between the bundle she has chosen and being able to choose from an *implicit* budget set with slope equal to $\tilde{w}$ and intercept equal to $m(\tilde{w}; y_i, l_i)$.⁷

The construction of the value of the index in Eq. (2) for individual i , given that she has chosen the bundle $z_i = (y_i, 1 - l_i)$, is shown in Fig. 2 in the space (h, y) , with $h \equiv 1 - l$. At z_i the welfare index for individual i is computed by identifying the bundle $z'_i = (y'_i, 1 - l'_i)$, which is the point of tangency between the indifference curve IC_i and the implicit budget line with reference slope $\tilde{w}$, and then by computing the value of the intercept $m(\tilde{w}, y_i, l_i)$. A greater satisfaction is necessarily associated with a greater implicit transfer $m(\tilde{w}, y_i, l_i)$ which is indeed a representation of i 's preferences.

In order to see that such a utility representation embodies the principle of compensation one can simply notice that, when comparing two individuals with equal preferences, the index always considers that the individual on a dominated indifference curve is worse off. This happens because the interpersonal comparison is performed by means of a common reference budget set and the final utility representation is neutral

⁷ A general normative characteristic of money-metric utilities is that they do not depend on the initial cardinalization U_i of individual preferences R_i , as the same value $m_i(\cdot)$ would be obtained with any increasing transformation of U_i (Fleurbaey and Blanchet 2013). As a consequence, the comparison of individual situations does not depend on the initial cardinalization of U_i .

with respect to the actual skills of individuals, in line with the compensation goal of eliminating inequalities due to skills.

As well established in the literature on responsibility-sensitive egalitarianism (Fleurbaey 2008), compensation minded indexes are not neutral with respect to preferences. This is where the choice of the parameter $\tilde{w}$ in $m_i(\tilde{w}, y_i, l_i)$ plays a role. If $\tilde{w}$ is low, individuals who are more work-averse tend to obtain lower implicit budgets than individuals who are less work-averse and hence a lower index (the contrary happens if $\tilde{w}$ is high). This is illustrated in Fig. 3. At a low value of $\tilde{w}$, individual a , who is more work-averse, is assigned a lower implicit budget. Indeed, when $\tilde{w} = \tilde{w}_2 < \tilde{w}_1$ one has $m_a(\tilde{w}_2, y_a, l_a) < m_b(\tilde{w}_2, y_b, l_b)$. Namely, individual a is considered to be worse off than individual b . When the reference wage is higher, work lovers tend instead to be assigned lower implicit budget lines: with $\tilde{w} = \tilde{w}_1$ individual b is ranked as worse off. A change in the reference wage $\tilde{w}$ reflects therefore a change in the policy maker's evaluation of individual preferences and redistributive concerns. The fact that the measurement of well-being depends on the choice of a parameter may seem arbitrary. However, one can always develop an ethical theory justifying the choice. The ethical values at stake are clear, and there are already some studies on the normative considerations that may induce the choice of certain reference values rather than others (Fleurbaey and Maniquet 2015). This type of analysis is beyond the scope of our paper. We are simply interested in understanding how sensitive a tax reform evaluation, based on welfare orderings, is with respect to the choice of the well-being measure, in particular of $\tilde{w}$. More in detail, in the empirical analysis we will pick three reference wages: zero, the average wage rate in 1986 and the maximum wage rate in 1986.

Let us conclude this section noting that Fleurbaey and Maniquet (2011) also use the ideas of fairness discussed so far to single out ways to aggregate such indexes. They show, echoing Blackorby and Donaldson (1988), that mild egalitarian requirements, such as convexity of the social welfare function, can be satisfied only with an absolute priority for the worst-off. Here, we refrain from aggregating the indexes and we exclusively focus on the overall distribution of individual measures of well-being and on how this is affected by a change in the fiscal system.⁸

4 The econometrics of a structural labor supply model

In order to empirically evaluate the impact of the 1986 tax reform on their well-being, we proceed in steps. First, we estimate married couples' preferences for consumption and leisure which are heterogeneous with respect to observed and unobserved characteristics. Second, we use the estimated preferences and the chosen bundles to compute

⁸ As a matter of fact, relying on the social welfare functions proposed by Fleurbaey and Maniquet (2011), just knowing the properties of the tax function one could assess the goodness of a reform. Of course, as explained in Fleurbaey and Maniquet (2006), the assessment of a tax-benefit system depends on the value judgments pertaining to the treatment of preference heterogeneity and they specifically discuss the optimality of an EITC-type of scheme. Their result, however, crucially relies on the fact that they use a social welfare function of the maximin type. Hence, in order to assess a reform and to know which part of the tax function should be changed first is sufficient to know the earnings of the worst-off individual.

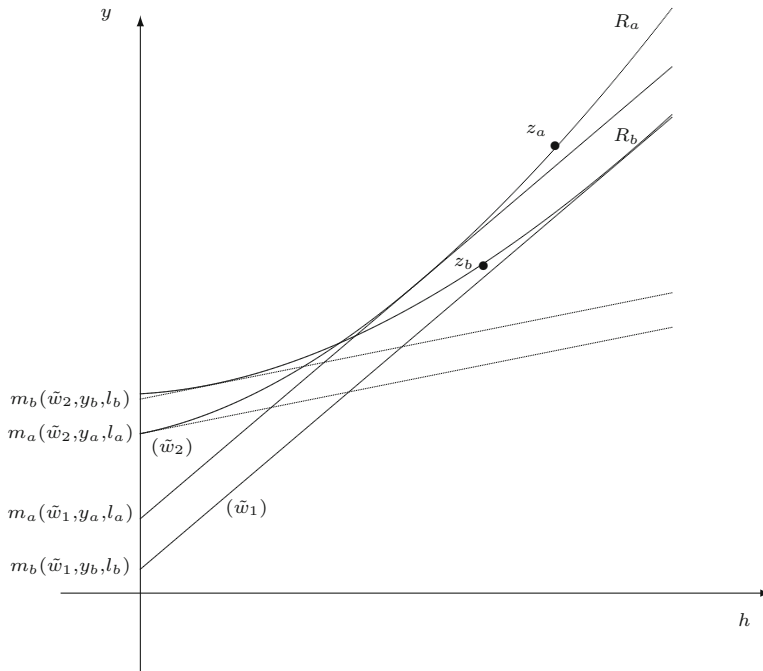


Fig. 3 How the choice of $\tilde{w}$ affects the measurement of individual well-being

the welfare metrics defined above. Finally, we simulate choices and compute welfare measures before and after the 1986 tax reform, so as to study its impact on the absolute and relative well-being of married couples.

4.1 Specification of household preferences

We estimate household preferences by means of a discrete choice structural labor supply model (van Soest 1995; Hoynes 1996; van Soest et al. 2002; Blundell and Shephard 2012). We focus on married or cohabiting women whose partners work at least 1500 h/year.⁹ Partners' labor supply is assumed to be exogenously determined. In the 1980s, women were typically secondary earners in the USA and empirical evidence shows that secondary earners are mostly affected by changes in the tax system, while primary earners are subject to a different set of incentives (Triest 1990; Blundell and Macurdy 1999).

Preferences are heterogeneous across households, and they depend on observed characteristics $\mathbf{x}$, unobserved characteristics $\mathbf{v}$ and a random term $\boldsymbol{\varepsilon}$. The direct utility of net income y and leisure time l is specified as $U(y, l; \mathbf{x}, \mathbf{v}, \boldsymbol{\varepsilon}) = u(y, l; \mathbf{x}, \mathbf{v}) + \varepsilon_l$. Like in Blundell and Shephard (2012), the random vector $\boldsymbol{\varepsilon}$, which is the additive

⁹ A similar sample selection can be found in the methodological contribution of van Soest et al. (2002), Eissa and Hoynes (2004), Bargain et al. (2013) and Decoster and Haan (2015).

discrete hours-specific error, is independent of $(\mathbf{x}, \mathbf{v})$ and the unobserved component $\mathbf{v}$ is independent of the observables $\mathbf{x}$.

The deterministic part of the direct utility function is assumed to have the following Box–Cox form

$$u(y, l; \mathbf{x}, \mathbf{v}) = \phi_y(\mathbf{x}_y, v_y) \frac{y^{\alpha_y} - 1}{\alpha_y} + \phi_l(\mathbf{x}_l, v_l) \frac{l^{\alpha_l} - 1}{\alpha_l}, \quad (3)$$

with $l \equiv 1 - h \in [0, 1]$, where h is wife's fraction of time devoted to work. The Box–Cox specification is often assumed in welfare analysis when estimating utility functions (Aaberge et al. 1995, 1999, 2004; Blundell and Shephard 2012; Bargain et al. 2013; Decoster and Haan 2015).¹⁰ The functions $\phi_y(\mathbf{x}_y, v_y)$ and $\phi_l(\mathbf{x}_l, v_l)$ determine the impact of observed and unobserved heterogeneity on preferences. They also determine the sign of the marginal utility and, jointly with the parameters α_y and α_l , the concavity of the deterministic part of the utility function. We restrict the power terms α_y and α_l to be between 0 and 1. Differently from Blundell and Shephard (2012), who imposed ϕ_y and ϕ_l to be positive, we parameterize them as linear functions in observed and unobserved characteristics:

$$\phi_y(\mathbf{x}_y, v_y) = \mathbf{x}'_y \boldsymbol{\beta}_y + v_y \quad (4)$$

$$\phi_l(\mathbf{x}_l, v_l) = \mathbf{x}'_l \boldsymbol{\beta}_l + v_l. \quad (5)$$

As proposed by van Soest (1995), labor supply is treated as a discrete choice problem.¹¹ We assume that each family can choose a bundle j of income and leisure combinations among 6 alternatives, i.e., $j = 1, \dots, 6$. Wives' yearly discrete working hours points, denoted by H , take values of (0, 390, 1020, 1536, 1976, 2500).¹² Each of these working hours points corresponds to the empirical median in the following working hours ranges: 0, 1–750, 751–1250, 1251–1750, 1751–2250 and more than 2250. Knowing the wage rate of each wife in our sample, one can derive the yearly gross income corresponding to each working hours choice and, after applying the tax-benefit function, we can derive the after tax family income for each H_j . To sum up, each married couple $i = 1, \dots, N$, when choosing a certain amount of leisure time l_{ij} with $j = 1, \dots, 6$, obtains the net income y_{ij} and the utility

$$U_{ij}(y_{ij}, l_{ij}; \mathbf{x}_i, \mathbf{v}_i, \boldsymbol{\varepsilon}_i) = u(y_{ij}, l_{ij}; \mathbf{x}_i, \mathbf{v}_i) + \varepsilon_{ij}. \quad (6)$$

We assume that the ε_{ij} 's follow an independent type I extreme value (EV-I) distribution. They are stochastic components capturing the fact that a given family could vary

¹⁰ See Dagsvik and Strøm (2006) for an evaluation of the performance of the Box–Cox functional form in analyzing labor supply behavior compared to less parametric (polynomial) specifications as studied by van Soest et al. (2002).

¹¹ See van Soest et al. (2002) for a discussion on the advantages and limits of the discrete labor supply model with respect to the continuous one.

¹² Leisure time enters the utility function through l , which is wife's fraction of time not devoted to work. More precisely, $l \equiv 1 - h = 1 - H/2500$. Hence, l is equal to 1 if $H = 0$ and is equal to 0 if $H = 2500$.

their tastes across bundles j (i.e., across different levels of working hours) because of unobserved attributes of the bundles which are independent across bundles. As pointed out by [van Soest \(1995\)](#), a way to interpret the ε_{ij} 's is family's errors in perception of the alternative utilities, for instance due to incomplete information or imperfect processing of the information ([Carpantier and Sapata 2016](#)), leading to optimization errors. Because of the independence assumption of the random terms across choice alternatives, they cannot be interpreted as reflecting heterogeneous preferences due to unobserved family characteristics ([van Soest 1995](#)). Although $\mathbf{v}_i$ is assumed to be independent of $\mathbf{x}_i$, i.e., it is a time-constant random effect at household level, it is able to capture heterogeneous preferences over consumption and leisure correlated across bundles due, for instance, to family characteristics that are not observed by the analyst.¹³

Each family is supposed to choose the bundle (y_{ij}, l_{ij}) for which U_{ij} is the largest. Under the type I extreme value distribution of the random terms $\boldsymbol{\varepsilon}$, the probability that family i chooses bundle j has a logistic form:

$$\Pr[U_{ij} > U_{ik}, \forall k \neq j | \mathbf{x}_i, \mathbf{v}_i] = \frac{\exp[u(y_{ij}, l_{ij}; \mathbf{x}_i, \mathbf{v}_i)]}{\sum_{k=1}^J \exp[u(y_{ik}, l_{ik}; \mathbf{x}_i, \mathbf{v}_i)]}. \quad (7)$$

The probability in Eq. (7) is the base to build the sample likelihood function and estimate the model by maximum likelihood (ML). However, we cannot use this density directly for ML estimation for two reasons. First, we need to know how y_{ij} is determined by l_{ij} for each $j = 1, \dots, 6$ and $i = 1, \dots, N$. Second, we do not observe $\mathbf{v}_i \equiv (v_{iy}, v_{il})$. The discussion about the determination of y_{ij} for each $j = 1, \dots, 6$ and $i = 1, \dots, N$ will be addressed in the next subsection. As far as the latter problem is concerned, we exploit the independence assumption between $\mathbf{v}_i$ and $\mathbf{x}_i$ and we integrate out $\mathbf{v}_i$ after imposing a bivariate discrete distribution ([Heckman and Singer 1984](#)) using a fixed number of support points with unknown locations and probability masses. We assume that v_y and v_l have both two points of support with corresponding probability masses defined as follows:

$$\begin{aligned} p_1 &\equiv \Pr(v_y = v_{1y}, v_l = v_{1l}) & p_2 &\equiv \Pr(v_y = v_{2y}, v_l = v_{2l}) \\ p_3 &\equiv \Pr(v_y = v_{2y}, v_l = v_{1l}) & p_4 &\equiv \Pr(v_y = v_{1y}, v_l = v_{2l}) = 1 - p_1 - p_2 - p_3. \end{aligned}$$

In this case, four points of support and three probability masses are to be estimated.¹⁴ The probabilities associated with the mass points are specified as logistic transforms:

$$p_m = \frac{\exp(\lambda_m)}{\sum_{r=1}^4 \exp(\lambda_r)} \quad \text{with } \lambda_4 = 0. \quad (8)$$

¹³ It is worthy to remark that the parametric assumption of the random component $\boldsymbol{\varepsilon}$ is not restrictive. [McFadden and Train \(2000\)](#) showed indeed that under mild regularity conditions any discrete choice model derived from random utility maximization can be approximated by EV-I errors in combination with a nonparametric specification of unobserved individual characteristics determining preferences.

¹⁴ Note that v_y and v_l are independent if and only if $p_1 p_2 = p_3 p_4$ ([van den Berg et al. 1994](#); [van den Berg and Lindeboom 1998](#)), making it easy to test for the independence of the unobserved components.

By integrating $\mathbf{v}_i$ out of density (7), we obtain the marginal density

$$\Pr[U_{ij} > U_{ik}, \forall k \neq j | \mathbf{x}_i] = \sum_{m=1}^4 p_m \frac{\exp[u(y_{ij}, l_j; \mathbf{x}_i, \mathbf{v}_{im})]}{\sum_{k=1}^J \exp[u(y_{ik}, l_k; \mathbf{x}_i, \mathbf{v}_{im})]}. \quad (9)$$

It is important to stress that the presence of $\mathbf{v}$ allows us to go beyond the assumption made by [Decoster and Haan \(2015\)](#) that two households with the same personal characteristics $\mathbf{x}$ would have the same preferences. As pointed out by [Carpantier and Sapata \(2016\)](#), the use of explicit unobserved heterogeneity at individual level and correlated across alternatives allows us to obtain a finer representation of individual preferences, which better matches households' actual choices, providing a more accurate assessment of individual situations.

4.2 Budget constraint and tax simulation

As said, we need to know how y_{ij} is determined by l_{ij} for each $j = 1, \dots, 6$ and $i = 1, \dots, N$. In the empirical analysis, we will use data from PSID from 1984 to 1990, which contains information on working time, income (labor income, social security benefits, welfare benefits, property income and pensions) and spouses' and family characteristics. In order to understand how wife's choice of working hours affect after tax family income, we need to:

1. Calculate wives' pre-tax hourly wage, so that we can compute yearly labor incomes for each discrete working hour point H . Using the information in the PSID dataset, we can compute the average hourly wage of each worker. However, if the wife is not working, the information is missing. For non-workers, we need then to predict the before tax wage rate. In a panel data setting, we estimate log wage equations for wives accounting for sample selection. Estimates of the wage equation are presented in Appendix A.2. The variables used to predict wives' wages are age, race, education, parents' education, time dummies and the unemployment rate in the state of residence.¹⁵ As suggested by [Hoynes \(1996\)](#), to maintain a consistent stochastic specification, predicted hourly wages from the estimated equation are used both for workers and non-workers. In order to properly take account of measurement errors in wages, we should have simultaneously estimated the log wage equation and the parameters of the utility function ([van Soest 1995](#); [van Soest et al. 2002](#); [Blundell 2001](#)). However, we derive the net family income from the gross income through an external tax simulator, the Tax Simulation Model (TAXSIM) of the National Bureau of Economic Research (NBER).¹⁶ Hence, we cannot integrate the tax-benefit simulation in each iteration of the ML estimation and we are forced, as in [Hoynes \(1996\)](#), [Bargain et al. \(2013\)](#) and [Decoster and](#)

¹⁵ In the selection equation, we include also the number of children younger than 17, an indicator for the presence of children younger than 6 and the number of family members.

¹⁶ We used the Internet Taxsim (v9). See [Feenberg and Coutts \(1993\)](#) for an introduction to TAXSIM. Further information can be found in Internet at <http://www.nber.org/taxsim/>. TAXSIM has already been used in the empirical studies of [Eissa et al. \(2008\)](#), [Bargain et al. \(2014\)](#) and [Bargain et al. \(2013\)](#).

Haan (2015), to stick to a reduced form method for imputing wages to non-workers which is distinct from the structural model of labor supply. As pointed out by van Soest (1995), using predicted wage rates for both workers and non-workers yields consistent estimates if all families based their choices on these predictions instead of actual wage rates.

2. Compute the family budget constraint per each discrete working hour point. The budget constraint of family i at each discrete $j = 1, \dots, 6$ is

$$y_{ij} = w_i H_j + E_i + V_i - T(w_i H_j, E_i, V_i, \mathbf{z}_i, t), \quad (10)$$

where w_i is wife's predicted hourly wage, E_i is husband's labor earnings, V_i is non-labor income (e.g., property income, interest, pensions, veteran benefits, social security benefits).¹⁷ $T(w_i H_j, E_i, V_i, \mathbf{z}_i, t)$ is the tax function which determines the tax (subsidy) that the family has to pay (receive) for the fiscal year t at both federal and state level. The tax-benefit schedule is a function of labor and non-labor income but also of $\mathbf{z}_i$ which is a vector of family characteristics. It includes the state of residence and the number of dependent children. We assume that all married couples file jointly the tax declaration. The function $T(w_i H_j, E_i, V_i, \mathbf{z}_i, t)$ is numerically calculated using TAXSIM.

4.3 Maximum likelihood estimation and identification

Equation (9) represents the contribution to the likelihood function of family i when it chooses H_j . In our empirical analysis, we will operate in a panel data framework. We assume that the bundle choices of each family are independent over time once we control for observed and unobserved characteristics determining preference heterogeneity. We also assume that families' preferences are time constant. So far, we have not indexed variables by the time indicator for the sake of clarity. We start now using the time subscript. Denote by g_{it} the variable indicating the bundle chosen by family i at time t and taking values of $\{1, \dots, 6\}$. The contribution to the likelihood function of family i is given by

$$\begin{aligned} \mathcal{L}_i &= \sum_{m=1}^4 p_m \left\{ \prod_{t=1}^T \left[\prod_{j=1}^6 \frac{\exp[u(y_{ijt}, l_j; \mathbf{x}_{it}, \mathbf{v}_{im})]}{\sum_{k=1}^6 \exp[u(y_{itk}, l_k; \mathbf{x}_{it}, \mathbf{v}_{im})]} \right]^{\mathbb{1}[g_{it}=j]} \right\} \\ &= \sum_{m=1}^4 p_m \left\{ \prod_{t=1}^T \left[\prod_{j=1}^6 \frac{\exp \left[\phi_y(\mathbf{x}_{it,y}, v_{im,y}) \frac{y_{ijt}^{\alpha_y} - 1}{\alpha_y} + \phi_l(\mathbf{x}_{it,l}, v_{im,l}) \frac{l_j^{\alpha_l} - 1}{\alpha_l} \right]}{\sum_{k=1}^6 \exp \left[\phi_y(\mathbf{x}_{it,y}, v_{im,y}) \frac{y_{itk}^{\alpha_y} - 1}{\alpha_y} + \phi_l(\mathbf{x}_{it,l}, v_{im,l}) \frac{l_k^{\alpha_l} - 1}{\alpha_l} \right]} \right]^{\mathbb{1}[g_{it}=j]} \right\}, \end{aligned} \quad (11)$$

¹⁷ We considered social security benefits as independent of wives' working hours choice. In each year of our longitudinal dataset, less than 1% of the families received social security benefits. From 1984 to 1990, only 2.4% of the families (61 out of 2529) received social security benefits in at least one year in which they were in the sample. We instead did not include in the budget constraint food stamps and the Aid to Families with Dependent Children declared in the PSID, since they depend on family income (and therefore on wives' working hours choice) but, differently from social security benefits, they are not handled by TAXSIM.

where $\mathbb{1}[\cdot]$ is the indicator function, equal to 1 when its argument is true. The log-likelihood function sums the logarithm of expression (11) over all the families and its parameters $\beta_y, \beta_l, \alpha_y, \alpha_l, \mathbf{v}_y, \mathbf{v}_l$ and $(\lambda_1, \lambda_2, \lambda_3)$ are estimated by ML.

The set of observed characteristics in both $\mathbf{x}_{it,y}$ and $\mathbf{x}_{it,l}$ are wife's and husband's age, a non-white indicator, wife's and husband's education (dummies for up to grade 8, from grade 9 to 11, grade 12, tertiary education without diploma and tertiary education with diploma) and indicators for the health condition of both spouses. The marginal utility with respect to leisure is supposed to be affected also by the presence of dependent children, so we included into $\mathbf{x}_l$ also the number of dependent children below 18 and a dummy indicator for the presence of children younger than 6. The variables for the presence of children, health and age are time-varying. Variables for education and race are time constant. See Table 11 in Appendix A.1 for summary statistics of the explanatory variables.

In our labor supply model, we impose a particular shape on the household utility function by assuming a Box–Cox form. This direct parametrization of the utility function certainly contributes to model identification. It is not, however, a necessary assumption for model identification. [van Soest et al. \(2002\)](#) have indeed shown how a structural labor supply model with a flexible nonparametric specification (series expansion in net income and hours) of the direct utility function can be used for the analysis of all sorts of tax-benefit changes. Moreover, in our framework, we exploit exogenous variation at different levels as further sources of model identification, leading us to a quasi-experimental set-up:

- Nonlinearities and discontinuities related to some family characteristics like the state of residence and the number of dependent children. This means that, at any point in time, families with the same before tax income but different $\mathbf{x}_{it}$ or different levels of non-labor income will face different tax rates and will thereby have different after tax income. Variation across states in income tax and EITC is particularly important, as pointed out by [Hoynes \(1996\)](#) and [Eissa and Hoynes \(2004\)](#), to the extent that geographical mobility is not correlated with work preference.
- Variations over time of the tax schedule with policy reforms both at state and federal level. Our longitudinal data cover the period from 1984 to 1990, and we can also take advantage of the important change in the tax schedule at federal level generated by TRA86.
- [Fox and Gandhi \(2016\)](#) show under what assumptions, with cross-sectional data, it is possible to nonparametrically identify the distribution of unobservables characterizing the heterogeneity among consumers in multinomial choice models. We are in a panel data setting and observe families multiple times in making bundle choices. This is of help in the nonparametric identification of the distribution of the unobserved characteristics that are constant over time.

5 Estimation results of the structural labor supply model

5.1 Data and sample

The data we use come from the 1984–1991 PSID.¹⁸ At that time, interviews were conducted annually on about 6,900–9,400 households each year. The PSID collects data on a wide set of social, demographic, economic and health topics. Information on income from several sources and working hours is retrospectively collected: information on income at time t is collected at time $t + 1$. Hence, our final dataset covers demographic, labor market and income information for the tax-years 1984–1990 and contains only families interviewed for at least two consecutive waves.

The sample includes married couples with the husband working at least 1500 h in a year and the wife between 20 and 55 years of age. We dropped families with one spouse working more than 4000 h in a year, with missing state of residence, with composition changes over time in the husband or wife position, with negative non-labor income and with one spouse retired, disabled, in jail or studying. The resulting sample is made up of 2542 different households (with more than 1900 couples interviewed per year), while the final sample used for the estimation of households preferences is made up of 2529 different families. After pooling all 7 years from 1984 to 1990, the resulting sample size is 13,275 households.¹⁹

5.2 Estimation results

The estimated parameters of the Box–Cox utility function in Eq. (3) are reported in Appendix A.3. As it is not easy to have a concrete interpretation of these parameters we rather employ them to compute, by way of microsimulations, the income variation needed to compensate a marginal variation in working hours (or leisure). The simulated marginal rate of substitution is computed per each household and then averaged across the pooled sample. We also estimate the average partial effect of selected characteristics on the marginal rate of substitution. The results, with and without unobserved heterogeneity v , are reported in Table 2.²⁰

Panel (a) of Table 2 shows that across our sample of married couples, a reduction of 50 h of work (increase of 50 h of leisure) of the wife must be compensated, in order to remain on the same indifference curve, by a decrease in the net income of \$494. The estimated average partial effects show that the reduction in net income is significant and large if there are children of preschooler age in the household and if the wife is in bad health. Also husband's bad health condition and wife's age have a significant and positive impact on distaste for work but smaller in size. When the wife turns one (ten) year(s) older, the marginal rate of substitution becomes larger in absolute value

¹⁸ The PSID is produced and distributed by the Institute for Social Research, Survey Research Center, University of Michigan, Ann Arbor, MI.

¹⁹ Further information on the size of the sample used for the estimation of households preferences is provided in Appendix A.1 together with descriptive statistics.

²⁰ We describe the simulation algorithm to obtain the figures in Table 2 in Appendix A.4.

Table 2 Marginal rates of substitution: yearly income variation to compensate a yearly decrease of 50 h of work in the model with and without unobserved heterogeneity $\mathbf{v}$

	Marginal rate of substitution (in \$1000)	95% confidence interval ^a	
<i>(a) With unobserved heterogeneity v</i>			
Whole sample	−0.4944	−0.7617	−0.3300
Average partial effects of selected characteristics on the marginal rate of substitution			
Age of wife (+1 year)	−0.0023	−0.0014	−0.0036
Age of wife (+10 years)	−0.0232	−0.0142	−0.0361
Age of husband (+1 year)	0.0014	0.0005	0.0027
Age of husband (+10 years)	0.0133	0.0050	0.0259
Number of kids below 18 years (+1)	0.0002	0.0000	0.0024
Presence of kids below 6 years	−0.0282	−0.0214	−0.0362
Non-white	0.0158	0.0046	0.0364
Wife with tertiary education with diploma	−0.0043	−0.0013	0.0775
Husband with tertiary education with diploma	0.0054	−0.0005	0.0270
Wife in bad health	−0.0410	−0.0170	−0.0864
Husband in bad health	−0.0063	−0.0249	0.0054
<i>(b) Without unobserved heterogeneity v</i>			
Whole sample	−0.3082	−1.0927	1.4552
Average partial effects of selected characteristics on the marginal rate of substitution			
Age of wife (+1 year)	−0.0043	−0.0061	−0.0027
Age of wife (+10 years)	−0.0437	−0.0601	−0.0278
Age of husband (+1 year)	−0.0000	−0.0015	0.0015
Age of husband (+10 years)	−0.0000	−0.0144	0.0149
Number of kids below 18 years (+1)	−0.0237	−0.0278	−0.0200
Presence of kids below 6 years	−0.1250	−0.1366	−0.1144
Non-white	−0.0113	−0.0266	0.0040
Wife with tertiary education with diploma	−0.2027	−0.2510	−0.1543
Husband with tertiary education with diploma	0.0654	0.0332	0.0982
Wife in bad health	−0.0057	−0.0370	0.0245
Husband in bad health	0.0033	−0.0191	0.0251

^a The 95% confidence intervals are computed from the predicted marginal rates of substitution by sampling 999 times the vector of parameter estimates assuming that the estimator is normally distributed around the point estimates with a variance–covariance matrix equal to the estimated one

by \$2 (\$23). Wives in non-white families, with an older husband and with a husband with a diploma of tertiary education, are instead more work lovers.

All in all, the marginal rates of substitutions are quite homogeneous across the different household types in our sample. This might be due to the unobserved heterogeneity $\mathbf{v}$ capturing variation in preferences across households. As a matter of fact, panel b) of Table 2 shows that if we did not take account of the presence of unobserved determinants correlated across alternatives: (i) we would have

underestimated the average marginal rate of substitution; (ii) we would have falsely attributed part of its variation to some observed determinants of preference heterogeneity.

Another way to understand what the estimated parameters of the structural labor supply mean in terms of labor supply behavior is to simulate uncompensated labor supply elasticities: we increase the net income by 1%, re-predict wives' labor supply and look at its change. We find that a 1% increase in net income implies an increase in working hours by 0.53% and in the employment rate by 0.19%.²¹ Our predicted labor supply elasticities are in line with other studies for the USA, especially with [Blau and Kahn \(2007\)](#) at the intensive margin and [Heim \(2009\)](#) at the extensive margin.²²

Finally, we have also checked the goodness of fit of the estimated model, i.e., its ability to replicate the actual choices made by the households in our sample. In general, the estimated model tends to overpredict the number of married wives who are not working and underpredict the number of wives working less than 750 h/year. We obtain a very good fit of the frequencies for each category of married women working more than 750 h/year. A detailed description of the results of the goodness-of-fit check is reported in Table 15 in Appendix [A.3](#).

5.3 Well-being indexes in 1986: actual and counterfactual predictions

We use the estimated parameters of the utility function to simulate the well-being level of the families in our sample in 1986, i.e., just before the application of TRA86, and compare it with the well-being level they would have reached in the counterfactual scenario where the new fiscal system is already operative. More in detail, for each family in the sample in 1986, we predict the 1986 net income–leisure choice by using, alternatively, the actual 1986 fiscal rules and the new ones implemented in 1988 and by keeping all the family characteristics fixed at the 1986 realizations. Then, we compute the well-being levels, using the three compensation minded metrics discussed above, attained by each family under the actual and the counterfactual scenario.²³

Table 3 reports summary statistics of the actual and counterfactual 1986 net income and of the three money-metric utilities. They all point to an average improvement of well-being induced by TRA86. In line with their theoretical construction, their cardinal value is decreasing on average in the reference wage rate $\tilde{w}$. By looking at the correlation among the three different welfare metrics, we find that they are highly correlated to the net income both in the actual and in the counterfactual scenario (correlation around 0.99). This is likely due to the fact that preferences over net income and leisure were not so heterogeneous in our sample although we find that individual

²¹ Further details and labor supply elasticities for different demographic groups are in Table 14, Appendix [A.3](#).

²² See also [Bargain et al. \(2016\)](#) for a recent survey on estimated labor supply elasticities in the USA and Western Europe.

²³ Appendix [A.5](#) describes the simulation algorithm to obtain the welfare metrics under the actual 1986 fiscal law and the counterfactual scenario with the 1988 fiscal rules.

Table 3 Summary statistics of the predicted 1986 welfare metrics and net income and counterfactual 1986 welfare metrics and net income using 1988 fiscal rules

Criterion (in \$1000)	Actual 1986				Counterfactual with 1988 fiscal rules			
	Mean	S.D.	Min.	Max.	Mean	S.D.	Min.	Max.
Net income	38.563	15.806	9.802	130.828	41.110	18.016	10.805	152.653
$\tilde{w} = 0$	34.679	15.553	6.199	125.484	36.969	17.625	7.084	146.168
$\tilde{w} = 1986 \text{ mean wage.}$	22.619	16.133	-6.577	115.731	24.959	18.224	-6.342	136.589
$\tilde{w} = 1986 \text{ max. wage.}$	2.719	16.316	-27.391	96.496	5.083	18.413	-27.154	117.424

The number of observations is equal to 1929, i.e., the number of households observed in our sample in 1986

characteristics are statistically highly significant in determining the profile of the utility function.²⁴

We also computed the natural logarithm of the position of each household in the ranking generated by each of the welfare metrics we use (and by net income) in 1986, and then we regressed it on the wage rate and the same set of characteristics entering the utility function. Table 4 reports the ordinary least squares (OLS) estimation results. The impact of the covariates on the 1986 ranking does not vary considerably across net income and the different welfare metrics, as expected given their high correlation. One more dollar of wage rate increases the ranking position (in terms of well-being) by about 18%. Wife's age is not correlated to the social ranking, while one more year in husband's age increases the position in the 1986 welfare rankings by about 21.5–23.4%. The presence of one more kid below 18 increases the position in the ranking by about 11%. The presence of kids younger than 6 does not affect the position in the ranking if one considers net income or the money-metric utility with $\tilde{w} = 0$. However, when the reference wage rate increases, a positive association pops up. When $\tilde{w}$ is equal to the mean wage, the presence of kids in preschooler age increases by 8.3%,²⁵ but not significantly different from zero, the position in the ranking. This figure becomes significant at 10% and equal to 9.9% with the reference wage equal to the maximum wage rate observed in 1986. This is consistent with the fact that with a high $\tilde{w}$, individuals that are more work-averse tend to obtain (relatively) higher values in the corresponding welfare metric (Fleurbaey and Maniquet 2015). Indeed, as shown earlier, the presence of kids in preschooler age increases the distaste for work.

Non-white households are worse off according to all the metrics, and their rank is about 30% lower than white households. Finally, education and health condition of wives do not play a role, whereas husbands' education strongly and positively affects the welfare ranking and the bad health condition of the husband decreases by about 25% the position in the social ordering.

²⁴ See Table 13 in Appendix A.4.

²⁵ $8.3\% = [\exp(0.080) - 1] \times 100$.

Table 4 OLS estimates of the log ranking equation in the welfare metrics in 1986

Variables	Money-metric utilities					
	Net income		$\tilde{u} = 0$		$\tilde{u} = 1986$ mean wage	
	Coeff.	S.E. ^a	Coeff.	S.E. ^a	Coeff.	S.E. ^a
Wage rate	0.212	***	0.180	0.026	0.182	0.025
Age of wife/10	0.026		0.012	0.059	0.051	0.057
Age of husband/10	0.226	***	0.234	0.052	0.223	0.051
Number of kids below 18	0.105	***	0.114	0.021	0.112	0.022
Presence of kids below 6	0.007		0.003	0.053	0.080	0.052
Non-white	-0.326	***	-0.346	0.049	-0.333	0.048
Education of wife—reference: up to grade 8						
Grades [9, 11]	-0.098		-0.095	0.158	-0.028	0.159
Grade 12	0.083		0.010	0.150	0.107	0.152
Tertiary without diploma	0.107		0.020	0.160	0.140	0.160
Tertiary with diploma	-0.290		-0.296	0.208	-0.175	0.203
Education of husband—reference: up to grade 8						
Grades [9, 11]	0.672	***	0.733	0.166	0.693	0.163
Grade 12	0.992	***	1.056	0.163	1.004	0.161
Tertiary without diploma	1.182	***	1.252	0.163	1.194	0.161
Tertiary with diploma	1.265	***	1.353	0.161	1.295	0.159
Wife in bad health	-0.004		0.005	0.067	0.016	0.065
Husband in bad health	-0.258	***	-0.253	0.091	-0.255	0.090
Constant	3.137	***	3.349	0.263	3.161	0.279
R^2	0.393		0.348		0.360	
# of households (N)	1929					

*** Significant at 1%; ** significant at 5%; * significant at 10%

a Standard errors are robust to heteroskedasticity

5.4 Absolute and relative gainers from TRA86

In this subsection we start by comparing, at household level, the well-being level attained in 1986 under the old fiscal system with the counterfactual well-being attained if the 1988 fiscal set-up was in place. We find that virtually all the households of our sample experienced an increase in well-being, regardless of the metric used (99.7%) and in net income (99.8%). About 1% of the households experiencing an increase in well-being also faced a reduction in their net income. This signals the importance of taking preferences into account in this kind of analysis. TRA86 reduced the labor supply of these married women so that their households experienced a decrease in the net income. However, they benefited from an overall increase in well-being because, given their preferences, they put more value on the extra leisure than on the foregone earnings. In terms of labor supply, we find that TRA86 had on average a positive effect on labor force participation: the fraction of time devoted to work increased by about 2% points, which amounts to an annual increase of 80 h of work.

The fact that virtually everybody wins, in terms of personal well-being, is due to the fact that TRA86 enlarged the budget constraint of the families in the sample, as also suggested by Fig. 1. It is hence more interesting to look at relative gains in the social ranking and at how these depend on the chosen welfare metric. To put it differently, we look now at what kind of households were able to climb the social ladder after the introduction of TRA86.

Table 5 reports descriptive statistics about the shuffle of households' positions in the well-being and in net income rankings when switching from the 1986 fiscal set-up to the 1988 tax system. Panel (a) of Table 5 displays the fraction of households climbing the social ladder thanks to TRA86. About 40% of the married couples have gained positions in the welfare ranking. This is roughly the same for the three welfare metrics we consider. If we disaggregate these statistics by selected percentiles for each different welfare metric distribution, we do not find noticeable differences.

Panel (b) of Table 5 focuses instead on the average number of positions gained by households in selected percentiles of each welfare metric distribution. On average, those at the bottom of the well-being distribution gain while those at the top lose if one looks at the overall effect of the reform. Very importantly, and this seems an essential aspect of the reform but overlooked by the economic literature, there has been a noticeable redistribution of net income and well-being between the lower-middle class (between the 10th and the 50th percentiles) and the upper-middle class (between the 50th and the 90th percentiles). Moreover, the lower the reference wage $\tilde{w}$, the stronger this redistribution. The reform can hence be considered to have been more or less redistributive depending on the chosen welfare metric.

What are the characteristics of the households climbing the social ladder thanks to TRA86? To answer this question, we computed for each household the ranking position in the income distribution and in the well-being distribution, for each welfare metric, before and after TRA86 and calculated its variation. We regressed the ranking position variation on the wage rate and the characteristics determining observed preference heterogeneity. Table 6 reports the OLS estimation results of the equation for the variation in the welfare ranking position. Some points are worth mentioning. First, both the signs and the magnitudes of the covariates considerably change across

Table 5 Gainers from TRA86 in terms of relative welfare ranking

	Net income	Money-metric utilities		Observations
		$\tilde{w} = 0$	$\tilde{w} = 1986 \text{ mean wage}$	$\tilde{w} = 1986 \text{ max. wage}$
(a) Fraction of households gaining positions in the ranking				
Overall	0.423	0.409	0.417	0.414
Selected percentiles of the 1986 welfare metric				
Between 0th and 10th percentiles	0.430	0.508	0.478	0.472
Between 10th and 50th percentiles	0.412	0.389	0.415	0.408
Between 50th and 90th percentiles	0.436	0.403	0.402	0.407
Between 90th and 100th percentiles	0.404	0.415	0.425	0.404
(b) Average position gain in the ranking				
Overall minimum and maximum ^a	[-51, 115]	[-52, 100]	[-52, 91]	[-57, 98]
Selected percentiles of the 1986 welfare metric				
Between 0th and 10th percentiles	0.238	0.352	0.067	0.155
Between 10th and 50th percentiles	0.592	0.364	0.368	0.153
Between 50th and 90th percentiles	-0.630	-0.401	-0.324	-0.140
Between 90th and 100th percentiles	-0.089	-0.207	-0.244	-0.207

Table 6 continued

Variables	Money-metric utilities					
	Net income		$\tilde{w} = 0$		$\tilde{w} = 1986$ mean wage	
	Coeff.	S.E. ^a	Coeff.	S.E. ^a	Coeff.	S.E. ^a
Education of husband— reference: up to grade 8						
Grades [9, 11]	−0.780	1.505	0.679	1.227	1.328	1.187
Grade 12	1.308	1.631	1.420	1.284	2.437	1.258
Tertiary without diploma	1.450	1.680	0.108	1.332	1.329	1.313
Tertiary with diploma	2.363	1.797	1.413	1.491	2.443	1.444
Wife in bad health	1.205	1.368	0.090	1.171	0.278	1.145
Husband in bad health	−2.425	*	1.383	0.239	−0.103	1.270
Constant	−7.809	**	3.563	−14.470	***	3.032
R^2	0.098		0.069	***	−10.338	2.957
# of households (N)	1929		0.065		0.065	0.060

*** Significant at 1%; ** significant at 5%; * significant at 10%

^a Standard errors are robust to heteroskedasticity

the different welfare metrics: it is not only important to take preferences into account, but also the weight which the policy maker gives to preferences matters.

Second, TRA86 favored people with higher wage rates regardless of which well-being index we consider but, interestingly, the effect of net income is not significant. The magnitude of this effect slightly increases when the reference wage rises. If $\tilde{w} = 0$, one more dollar of hourly wage translates into 1 position gained in the welfare ranking (out of 1929 observations) with the introduction of TRA86. If $\tilde{w}$ is equal to the maximum wage rate, one more dollar of the wage rate would imply a gain of 1.2 positions.

Third, while according to all welfare metrics (and to net income) the presence of kids below 18 has a positive effect on the ranking position, essentially because of the larger generosity of the EITC, the presence of kids in preschooler age differently affects the variation in the position in the welfare ranking depending on the welfare metric. When $\tilde{w}$ is equal to the mean or the maximum wage rate, the presence of a kid below 6 reduces the ranking position by 2–2.5 positions after the introduction of TRA86. When $\tilde{w} = 0$, the impact of the presence of kids in preschooler age is instead mildly positive and equal to about 0.3 positions (the impact of the presence of kids in preschooler age is even weaker if one considers net income). As we have seen earlier, when the reference wage is higher, work-averse individuals tend to be assigned to a higher implicit budget line and hence to a higher well-being. As Table 2 shows, women with kids younger than 6 are more work-averse. Hence, with a high $\tilde{w}$ and everything else constant, they were already ranking high before the reform compared to households without kids in preschooler age. They are therefore more likely to lose positions in the ranking with the introduction of the reform. This is true, even if to a lesser extent, also for other characteristics that have a positive impact on the work aversion.

Lastly, non-white households lose position regardless of the chosen welfare metric. The loss is even stronger considering the net income. Families with highly skilled wives have considerably lost in relative terms, especially when $\tilde{w}$ is equal to the mean or the maximum wage rate or if one considers net income. As we are considering only households with a husband working full time, the presence of a highly educated woman is consistent with the fact that such a household has benefited less from the EITC increase as compared to other families.

5.5 Decomposing the effect in income and behavioral effects

The change in the tax system produces both a direct income effect on the utility level that can be attained by the household, due to the fact that the budget constraint changes, and a behavioral effect, due eventually to a change in labor supply in response to the reform. Similar to [Creedy and Héroult \(2015\)](#), we decompose the reform effect in terms of individual relative welfare gains, into a direct income effect, due to tax and transfer changes, and a behavioral effect, due to wives' labor supply responses. Our decomposition is graphically illustrated in Fig. 4. Assume that agent i chooses the bundle $z_i = (y_i, 1 - l_i)$ before the reform and the bundle $z_i'' = (y_i'', 1 - l_i'')$ after the reform. The change in well-being is equal to $m_i(\tilde{w}, y_i'', l_i'') - m_i(\tilde{w}, y_i, l_i)$, which is

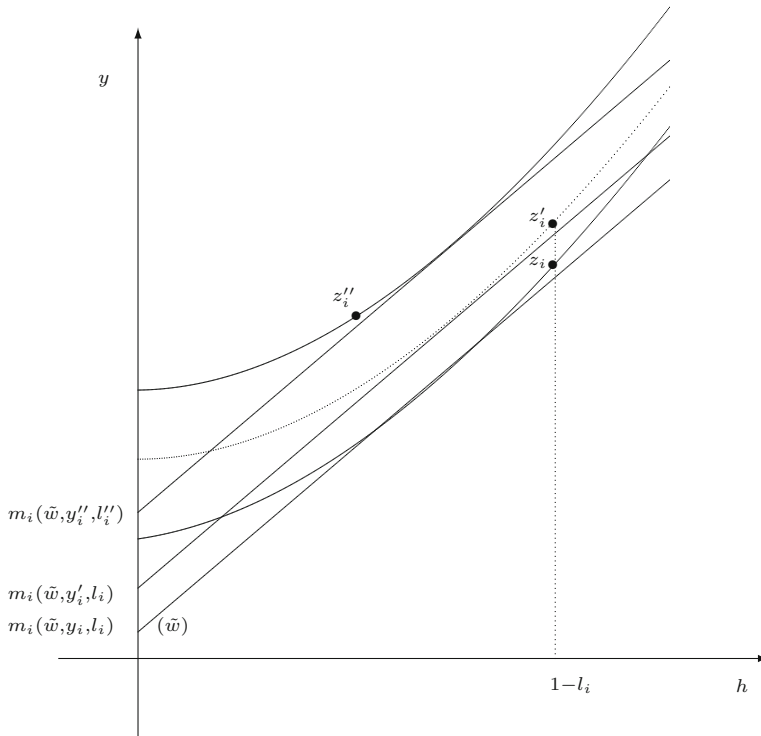


Fig. 4 Decomposition of the well-being differential determined by the reform

positive in this example. We first consider the change in the money-metric utility purely due to the change in the fiscal regime: the difference between the utility $m_i(\tilde{w}, y_i', l_i)$, associated with the bundle $z_i = (y_i', 1 - l_i)$ obtained fixing the labor supply of agent i to her pre-reform level $1 - l_i$ and giving her the net income y_i' granted by the new tax code at $1 - l_i$, and the initial utility $m_i(\tilde{w}, y_i, l_i)$. We let then agent i choose, from the post-reform budget set, the bundle that maximizes her utility, $z_i'' = (y_i'', 1 - l_i'')$ and we compute the utility associated with such bundle, $m_i(\tilde{w}, y_i'', l_i'')$. The utility variation due to agent i 's behavioral response is then $m_i(\tilde{w}, y_i'', l_i'') - m_i(\tilde{w}, y_i', l_i)$.

The decomposition allows us to look at Tables 5 and 6 from a different and more interesting perspective. In Table 7, the average position gain is decomposed, for net income, for each welfare metric and for each percentile, into the average position gain due to the direct income effect and the average position gain due to the behavioral response. Three points are worthy of mention. First, the magnitude of the components increases with $\tilde{w}$. This signals the crucial role of the overall shape of the indifference maps, and hence of preference heterogeneity, not only in the calculation of the welfare level, but also for the welfare difference. It is consistent with the fact that at $\tilde{w} = 0$ the well-being is measured at $1 - l_i = 0$, the vertical intercept, where the marginal rates of substitution are more similar across heterogeneous individuals. When $\tilde{w}$ is equal to the maximum wage, the well-being is measured at a point where the marginal rates of

substitution differ more significantly across heterogeneous individuals and, therefore, the distance between them in terms of welfare metric is larger.

Second, the signs of average position gain due to the income components are the same as those of the total effect, while the behavioral components tend to compensate the income component. The fact that the sign of the behavioral effect on the average position gain is negative for households below the median of the welfare metric might be explained by the fact that these households weakly responded to the tax reform in terms of working hours.

Third, as mentioned earlier, by looking at the total effect of the reform on the social ranking (panel (b) of Table 5), one could conclude that the reform could be considered to be more redistributive, in terms of well-being, if one relies on money-metric utilities with $\tilde{w} = 0$. Interestingly, if one only looks at the income effect solely, then the reform could be considered to be more redistributive if one evaluates it at $\tilde{w}$ equal to the maximum wage.

To sum up, households at the bottom of the well-being distribution lost positions in the social ranking likely because of their weak behavioral response, while they mostly gained in terms of income effect, likely because of the EITC expansion. Moreover, the behavioral component compensates the income component in an increasing fashion: this is due to the fact that the different well-being measures not only give a different weight to leisure in general, but the way they do so depends on the shape of the entire indifference map.

Table 8 reports the OLS estimates of the equations for the change in the positions in the welfare ranking from TRA86 due to the direct income effect and due to the behavioral effect. When looking at the income effect, the dependent variable is the change in the position in the social ranking if wives were forced to stick to the same working hours as before the reform, with households being taxed using the 1988 system. When focusing on the behavioral component, the dependent variable is the change in the position in the welfare ranking when wives are free to choose a new optimizing bundle with the new fiscal system. One should notice that the estimated coefficients reported in Table 6 are equal to the sum of the coefficients of the two equations in Table 8.

The estimated coefficients from the equation for both the income and the behavioral effects change considerably across net income and money-metric utilities in general. This is true also within the latter indicators when the reference wage $\tilde{w}$ varies, especially when moving away from $\tilde{w} = 0$. Moreover, when $\tilde{w}$ is equal to the mean and the maximum wage rate, the sign of the income coefficient of a particular characteristic is systematically positive (negative) if the same characteristic has a positive (negative) impact on distaste for work. The opposite happens for the behavioral coefficients (compare Tables 2 and 8). This is consistent with the idea that, as $\tilde{w}$ grows, households that were treated more favorably by the well-being indexes, as more work-averse, lose positions in the social ranking because of the income effect. A positive behavioral effect systematically compensates this loss. The magnitude of the coefficients also tends to increase with $\tilde{w}$, once again this is consistent with the fact that at a higher $\tilde{w}$ the utility is measured at a point where the distance between indifference curves is bigger. This determines bigger utility differentials and bigger jumps in the well-being ranking.

Table 7 Decomposing the average position gain in the welfare ranking in income and behavioral effects

	Net income		$\tilde{w} = 0$		$\tilde{w} = 1986$ mean wage		$\tilde{w} = 1986$ maximum wage		Observ.
	Income effect	Behavioral effect	Income effect	Behavioral effect	Income effect	Behavioral effect	Income effect	Behavioral effect	
(a) Average position gain in the welfare ranking									
Overall min. and max. ^a	[-48, 89]	[-34, 44]	[-52, 96]	[-7, 10]	[-101, 125]	[-91, 96]	[-129, 141]	[-122, 118]	1929
Selected percentiles of the 1986 welfare metric									
Between 0th and 10th	0.264	-0.026	0.358	-0.005	1.342	-1.275	1.777	-1.622	193
Between 10 and 50th	0.313	0.278	0.333	0.031	0.937	-0.569	1.155	-1.003	772
Between 50 and 90th	-0.358	-0.272	-0.372	-0.029	-1.182	0.857	-1.505	1.364	771
Between 90 and 100th	-0.088	0.000	-0.202	-0.005	-0.368	0.124	-0.389	0.181	193

^a We do not report the average of the overall average gain in positions in the welfare ranking since it is 0 by construction

Table 8 continued

Variables	Money-metric utilities					
	Net income		$\tilde{w} = 0$		$\tilde{w} = 1986$ mean wage	
	Coeff.	S.E. ^a	Coeff.	S.E. ^a	Coeff.	S.E. ^a
Education of husband—reference: up to grade 8						
Grades [9, 11]	-2.125 **	0.827	-0.035	0.137	-11.952 ***	1.620
Grade 12	-1.233	0.822	0.015	0.135	-15.261 ***	1.542
Tertiary without diploma	0.256	0.847	0.043	0.144	-19.397 ***	1.614
Tertiary with diploma	-0.584	0.864	0.069	0.151	-19.979 ***	1.758
Wife in bad health	0.501	0.550	0.221	0.108	3.128 **	1.354
Husband in bad health	-1.413 **	0.599	-0.204	0.104	0.340	1.414
Constant	0.358	1.672	0.809	0.314	-47.979 ***	4.112
# of households (<i>N</i>)	1929				-33.058 ***	6.516

*** Significant at 1%; ** significant at 5%; * significant at 10%

^a Standard errors are robust to heteroskedasticity

The positive sign of the behavioral coefficient of a given characteristic is consistent with the fact that an increase in that characteristic makes the wife more likely to modify her labor supply after the reform and to gain positions in the ranking because of this change. On the contrary, a negative sign of the behavioral coefficient of a given regressor may signal that an increase in that regressor is not likely to affect at the margin the labor supply and make, therefore, wives more likely to stick to the pre-reform labor supply. When the behavioral response is weak, *ceteris paribus*, a household loses positions in the welfare ranking. It is instead more difficult to understand the direction of the labor supply change in case of a positive behavioral effect. For example, families with kids below 6 might have had a negative behavioral response (working fewer hours) as they are more work-averse and they have benefited more from the EITC. It seems more plausible that highly educated women have decided to work more as they may have benefited less from the EITC increase.

6 Conclusions

In this paper, we evaluated the effect of TRA86, the US fiscal reform of 1986, on the well-being of married couples. First, we estimated a structural model of labor supply in which households have preferences over net income and leisure that are allowed to be heterogeneous in observed and unobserved characteristics. Second, we used the estimated preferences to predict the optimal bundles of leisure and net income for 1986 households before TRA86 and in the counterfactual scenario where TRA86 was in place. Third, on the basis of households' choices under the actual and counterfactual scenarios, we computed the well-being level of each household using several indexes, as proposed by [Fleurbaey and Maniquet \(2011, 2015\)](#), which are based on different ethical priors, i.e., they reflect different value judgments the social planner may want to put forward. Finally, we analyzed how many and which type of households experienced a relative increase in well-being and how the change of their position in the welfare ranking was related to a change in income generated by the reform or to households' behavioral response to the reform.

We conclude that the introduction of TRA86 increased the well-being of almost every couple in our sample. However, depending on the normative considerations behind the different ways of measuring well-being, we arrive at different verdicts about what types of households gained most, relative to the other members of society. We observe that TRA86 has induced, within our sample, a redistribution of income and well-being between the lower-middle class and the upper-middle class. However, the strength of the redistribution clearly depends on the well-being index (and the corresponding value judgment) one uses to assess the reform. This is more transparent if one considers separately the well-being variation solely due to the income change determined by the reform and the well-being variation due to the behavioral response of individuals.

By selecting married couples with the husband working more than 1500 h/year, we oversampled relatively richer couples.²⁶ It would be interesting to see whether the

²⁶ More details are provided in Appendix [A.1](#).

reform had comparable effect on the distribution of well-being of single mothers and of couples with the husband not working full time, populations that over-represent low-income households. This is left to future research.

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Appendix

A.1 More information about the data and sample

Column (1) of Table 9 displays the number of household observations in the original PSID dataset, while column (2) reports the number of observations after applying our selection criteria. The sample in column (2) is the one used to predict wives' hourly wages. After predicting wives' hourly wages and simulating the net income which corresponds to the discrete hour point of the interval actually chosen, we removed from the sample households in the last percentile of the pooled net income distribution (9 households removed). We also deleted households with nil net income when the wife is supposed not to work (4 households removed). Finally, column 3 provides detailed information about the size of the sample used for the estimation of households preferences.

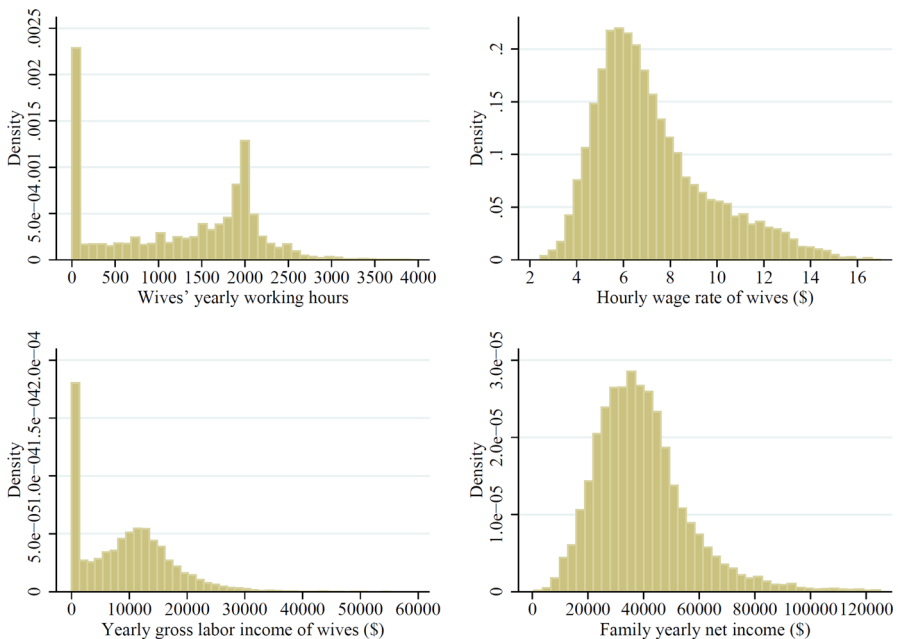
Table 10 reports the distribution of wives' working hours. About 21% of wives in our sample do not work. The remaining married women are almost equally split into those working full time (more than 1750 h/year, in average 35 h/week) and those working part-time (less than 1750 h/year). The top-left panel of Fig. 5 displays the

Table 9 Number of household observations from the original PSID dataset to the final selected sample

Years	Original PSID sample (1)	Sample used for predicting wives' hourly wages (2)	Final sample after removing outliers in predicted net consumption (3)
1984	6918	1953	1934
1985	7032	1938	1918
1986	7018	1945	1929
1987	7061	1917	1897
1988	7114	1874	1849
1989	7114	1896	1864
1990	9371	1912	1884
# of different families (<i>N</i>)	11,774	2542	2529
# of pooled observations (<i>NT</i>)	51,628	13,435	13,275

Table 10 Working hours distribution of married women

Yearly working hours (h)	Absolute frequency	Relative frequency	Mean	Median
0	2746	0.207	0.000	0.000
(0, 750]	1732	0.131	386.912	390.000
(750, 1250]	1428	0.108	1013.692	1020.000
(1250, 1750]	2050	0.154	1528.397	1536.000
(1750, 2250]	4354	0.328	1978.260	1976.000
(2250, 4000]	965	0.073	2589.285	2500.000
Total	13,275	1.000	1232.610	1476.000

**Fig. 5** Distribution of wives' working hours, their wage rates, yearly gross labor income and family net income

working hours distribution. The top-right panel is instead the distribution of married women's predicted hourly wages. The bottom panels of Fig. 5 focus on wives' gross labor income and family yearly after tax income.²⁷ Table 11 reports the descriptive statistics of the variables used in the estimation of the utility function. Married women are on average 35.5 years old, 2.5 years younger than their husbands. There are 1.5 dependent children (17 years old or younger) in each married couple, and in 68% of the couples there is a child younger than 6. One quarter of the couples are non-white.

²⁷ The net income is deflated to 1990 prices using the consumer price index by the US Bureau of Labor Statistics and available at <http://ftp.bls.gov/pub/special.requests/cpi/cpiiai.txt>.

Table 11 Summary statistics

	Mean	SD	Minimum	Maximum
Predicted family net income (\$)	39,416.84	16,737.61	673.3	125,114.0
Age of wife	35.538	8.031	20	55
Age of husband	37.968	8.690	20	67
Number of kids below 18 years	1.473	1.132	0	4
Presence of kids below 6 years	0.680	0.864	0	4
Non-white	0.246	0.430	0	1
Education of wife				
Less than grade 9	0.032	0.177	0	1
Grades [9, 11]	0.115	0.319	0	1
Grade 12	0.437	0.496	0	1
Tertiary education without diploma	0.228	0.420	0	1
Tertiary education with diploma	0.187	0.390	0	1
Education of husband				
Less than grade 9	0.048	0.214	0	1
Grades [9, 11]	0.131	0.337	0	1
Grade 12	0.346	0.476	0	1
Tertiary education without diploma	0.228	0.420	0	1
Tertiary education with diploma	0.247	0.431	0	1
Wife in bad health	0.089	0.285	0	1
Husband in bad health	0.074	0.261	0	1
# of pooled observations (<i>NT</i>)	13,275			

Almost 9% of the married women declare to be in bad health, while the same figure for men is about 7.4%. Married men are more likely to have a diploma of tertiary diploma (24.7%) than their wives (18.7%). However, married women are less likely than married men to have a low educational attainment: while 17.9% of the husbands stopped at grade 11 or less, only 14.7% of married women stopped at grade 11 or less.

In comparing the distribution of these demographic characteristics between husbands and wives, it must be reminded that we selected families with the husband working at least 1500 h/year and the wife between 20 and 55 years of age. These two selection criteria are the most important ones in terms of number of observations removed from the original sample of married couples. In order to shed more light on the consequences on the representativeness of the selected sample of married couples in terms of family income, in Fig. 6 we plot the kernel density estimates of the family yearly net incomes of: (i) our final sample of married couples (in blue); (ii) the whole sample of married couples including also those with the husband working less than 1500 h/year and with the wife older than 55 or younger than 20 (in green); (iii) the whole sample of married couples but excluding those with the wife older than 55 or younger than 20 (in red). The vertical lines indicate the average of each distribution. Figure 6 shows that the selection of families with the husband working

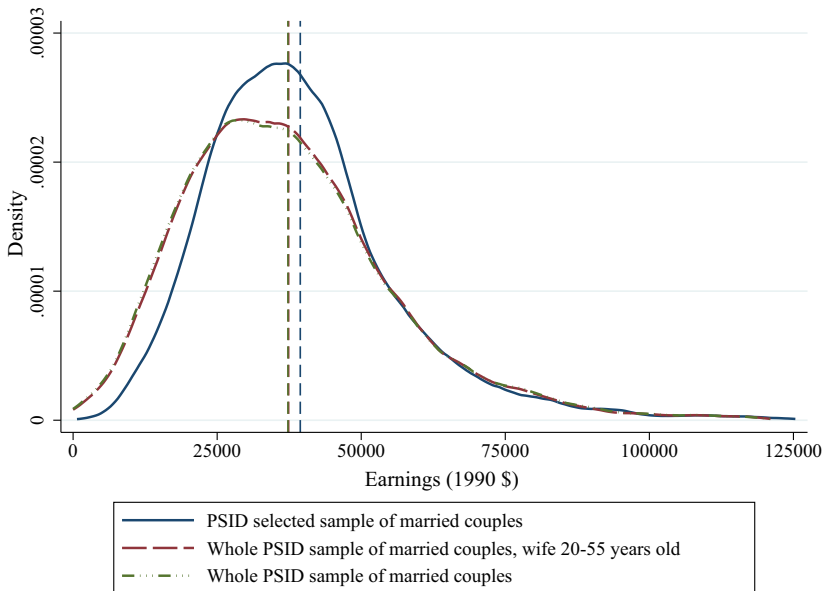


Fig. 6 Kernel density estimates of the family yearly net incomes for our selected sample of married couples (blue), for the whole sample of married couples (green), for the whole sample excluding married couples with wife younger than 20 or older than 55 (red). Notes: the vertical dashed lines indicate the mean of the household earnings of each distribution. We used the Epanechnikov kernel in calculating the kernel density estimates (Color figure online)

more than 1500 h/year generated the oversampling of relatively richer couples. The age restriction had instead no influence. (Color figure online)

A.2 Sample selection corrected estimation of log wage equation

In order to account for non-workers' missing wages, log wage equations for the wives are estimated taking into account sample selection bias. We first tested for sample selection following Wooldridge (1995).²⁸ The log wage equation is

$$\ln w_{it} = \mathbf{x}_{it,w} \boldsymbol{\beta}_w + c_i + u_{it}, \quad t = 1, \dots, T, \quad (\text{A.1})$$

where w_{it} is observed only if wife i is working, $\mathbf{x}_{it,w}$ is the set of exogenous regressors determining wages at time t , c_i is time-invariant unobserved characteristics and u_{it} is an idiosyncratic error term. Denote by s_{it} the binary indicator equal to 1 if woman i is working at time t and 0 otherwise. Assume that, for each t , s_{it} is determined by the following probit model

$$s_{it} = \mathbb{1}[\mathbf{x}_{it,s} \boldsymbol{\beta}_s + \bar{\mathbf{x}}_{i,s} \boldsymbol{\delta}_s + v_{it}], \quad v_{it} | \mathbf{x}_{it,s} \sim \text{Normal}(0, 1), \quad (\text{A.2})$$

²⁸ The notation in this section is inspired by the one in Wooldridge (2010, Sect. 19.9.2).

where $\mathbf{x}_{it,s}$ contains $\mathbf{x}_{it,w}$ and $\bar{\mathbf{x}}_{i,s}$ is the average of $\mathbf{x}_{it,s}$ over t and represents the Mundlak (1978) approach to allow the unobservables to be correlated with some elements of the observed characteristics in a discrete choice panel data model.

The parameters of the model in Eq. (A.2) are estimated by pooled probit and reported in the first columns of Table 12. We can test for sample selection by computing the inverse Mills ratio obtained from the pooled probit estimation and by plugging it into the log wage equation as a further regressor. Under the null assumption of no sample selection, the inverse Mills ratio should not be significant when the log wage equation is estimated by fixed effects (FE). The bottom line of Table 12 displays a Wald test for the significance of the inverse Mills ratio after the FE estimation of the log wage equation. The test is made robust to heteroskedasticity and serial correlation. As the inverse Mills ratio is highly significant, a problem of sample selection is detected and should be corrected for.

In order to correct for sample selection, we run pooled ordinary least square regression on the log wage equation augmented by the inverse Mills ratio, its interaction with the time dummies and $\bar{\mathbf{x}}_{i,s}$. The estimation results of the coefficients of the log wage equation are reported in the last three columns of Table 12. These estimated coefficients are used to predict wages for both working and non-working wives.

A.3 Estimated parameters, labor supply elasticities and goodness of fit

Table 13 reports the estimation results of the Box–Cox utility function in Eq. (3). We find that it is important to account for the presence of unobserved heterogeneity²⁹: the log-likelihood function improves from $-22,291.6$ without unobserved heterogeneity to $-18,052.4$ with unobserved heterogeneity specified as explained in Sect. 4.1. We also find that the unobserved components determining preferences in leisure and the unobserved ones determining preferences in income are positively and significantly correlated.

Since we constrained α_l and α_y to be between 0 and 1, the marginal utilities are positive and the concavity conditions are satisfied if and only if $\phi_y(\mathbf{x}_y, v_y) > 0$ and $\phi_l(\mathbf{x}_l, v_l) > 0$. The utility function and its first and second derivatives depend, however, not only on observed characteristics but also on unobservables. Since we have estimated the distribution function of the unobservables, we can derive the mixed utility function and check whether the corresponding first and second derivatives satisfy the monotonicity and the concavity conditions. The estimated mixed utility function is

$$\sum_{m=1}^4 \hat{p}_m u(y, l; \mathbf{x}, \hat{\mathbf{v}}_m) = \sum_{m=1}^4 \hat{p}_m \hat{\phi}_y(\mathbf{x}_y, \hat{v}_{my}) \frac{y^{\hat{\alpha}_y} - 1}{\hat{\alpha}_y} + \sum_{m=1}^4 \hat{p}_m \hat{\phi}_l(\mathbf{x}_l, \hat{v}_{ml}) \frac{l^{\hat{\alpha}_l} - 1}{\hat{\alpha}_l}. \quad (\text{A.3})$$

²⁹ We do not report the estimation result of the model without unobserved heterogeneity for the sake of brevity. They are available from the authors upon request.

Table 12 Estimated parameters of the employment selection equation and sample selection corrected log wage equation

Variables	Probit selection equation		Corrected log wage equation	
	Coeff.	SE	Coeff.	SE ^a
Constant	0.957	0.311	0.301	0.263
Age	0.051	0.015	0.085	0.013
Age square	-0.001	0.000	-0.001	0.000
Race—reference: white				
Black	0.245	0.034	0.020	0.033
Other	0.253	0.082	0.032	0.071
Education of wife—reference: up to grade 5				
Grades [6, 8]	-0.454	0.143	-0.276	0.134
Grades [9, 11]	-0.270	0.128	-0.247	0.103
Grade 12	0.100	0.126	-0.135	0.095
Tertiary education without diploma	0.271	0.128	0.019	0.099
Tertiary education with diploma	0.451	0.130	0.335	0.102
Education of father—reference: up to grade 5				
Grades [6, 8]	-0.027	0.049	0.014	0.044
Grades [9, 11]	-0.025	0.059	0.054	0.052
Grade 12	0.022	0.054	0.088	0.046
Tertiary education without diploma	-0.026	0.070	0.112	0.066
Tertiary education with diploma	-0.092	0.068	0.133	0.063

Table 12 continued

Variables	Probit selection equation		Corrected log wage equation	
	Coeff.	SE	Coeff.	SE ^a
Education of mother—reference: up to grade 5				
Grades [6, 8]	0.239	0.061	0.045	0.066
Grades [9, 11]	0.126	0.063	−0.004	0.068
Grade 12	0.137	0.062	0.077	0.065
Tertiary education without diploma	0.032	0.077	0.055	0.076
Tertiary education with diploma	0.059	0.081	0.007	0.079
Time dummies—reference: 1984				
1985	−0.051	0.049	0.039	0.044
1986	−0.061	0.054	0.097	0.050
1987	−0.029	0.064	0.124	0.055
1988	0.024	0.076	0.191	0.060
1989	0.080	0.087	0.281	0.066
1990	−0.005	0.099	0.310	0.074
State unemployment rate	2.647	1.670	−1.306	0.789
Number of kids below 18 years	−0.015	0.046	—	—
Presence of kids below 6 years	−0.281	0.050	—	—
Number of household members	−0.070	0.044	—	—
Time average of age square	−0.000	0.000	−0.000	0.000

Table 12 continued

Variables	Probit selection equation		Corrected log wage equation	
	Coeff.	SE	Coeff.	SE ^a
Time average of state unemployment rate	-9.856	1.861	0.295	1.251
Time average of the number of kids below 18 years	-0.131	0.058	-0.100	0.033
Time average of presence of kids below 6 years	-0.430	0.066	0.121	0.061
Time average of the number of household members	0.107	0.055	-0.002	0.029
Inverse mills ratio	-	-	-0.261	0.200
Inverse mills ratio* 1985	-	-	0.108	0.128
Inverse mills ratio* 1986	-	-	-0.032	0.140
Inverse mills ratio* 1987	-	-	0.010	0.151
Inverse mills ratio* 1988	-	-	-0.098	0.153
Inverse mills ratio* 1989	-	-	-0.351	0.172
Inverse mills ratio* 1990	-	-	-0.280	0.163
Log-likelihood	-6429.3		-10,416.8	
# of parameters	35		39	
Pseudo R ² /R ²	0.072		0.195	
<i>NT</i>	13,435		10,597	
<i>N</i>	2542		2305	
<i>F</i> -test for sample selection	<i>F</i> (1, 2304) = 17.42, <i>p</i> -value = 0.000			

*** Significant at 1%; ** significant at 5%; * significant at 10%

^a The standard errors are computed by bootstrapping the results 1000 times (individual-cluster bootstrap with replacement)

Table 13 Estimated parameters of the Box–Cox utility function with unobserved heterogeneity

Variables	Leisure		Net income	
	Coeff.	Std. Err.	Coeff.	Std. Err.
Age of wife/10	0.446	***	−0.007	0.025
Age of husband/10	0.023		0.051	0.024
Number of kids below 18 years/10	−0.049		−	−
Presence of kids below 6 years	0.865	***	−	−
Non-white	0.381	***	0.118	0.024
Education of wife—reference: up to grade 8				
Grades [9, 11]	0.490	*	0.054	0.066
Grade 12	1.720	***	0.410	0.062
Tertiary education without diploma	2.458	***	0.516	0.066
Tertiary education with diploma	3.884	***	0.703	0.073
Education of husband—reference: up to grade 8				
Grades [9, 11]	−0.023		0.079	0.059
Grade 12	−0.228		0.034	0.054
Tertiary education without diploma	−0.264		0.050	0.056
Tertiary education with diploma	−0.472	*	−0.049	0.056
Wife in bad health	−0.435	***	−0.195	0.037
Husband in bad health	−0.122		−0.040	0.034
α_l	0.487	***	−	−
α_y	−		0.859	0.017

Table 13 continued

Variables	Leisure		Net income	
	Coeff.	Std. Err.	Coeff.	Std. Err.
Unobserved heterogeneity support points				
v_1	-1.388	0.396	-0.174	0.094
v_2	3.642	0.403	2.040	0.160
Unobserved heterogeneity probability weights				
λ_1	0.459	0.062		
λ_2	0.155	0.062		
λ_3	-2.135	0.137		
λ_4	0.000	—		
Resulting unobserved heterogeneity probability masses ^a				
p_1	0.409	0.007		
p_2	0.302	0.007		
p_3	0.031	0.001		
p_4	0.258	0.011		
Log-likelihood	-18,052.4			
# of parameters	37			
AIC/NT	14,306			
NT	13,275			
N	2529			
Corr(v_y, v_l)	0.494			
LR test for independence of v_y and v_l , $H_0: v_y \perp v_l$	$\chi^2(1) = 498.6$, p -value = 0.000			

*** Significant at 1%; ** significant at 5%; * significant at 10%

^a The standard errors of the unobserved heterogeneity probability masses are estimated by the delta method

Hence, the monotonicity and concavity conditions are satisfied if, for each $i = 1, \dots, N$ and $t = 1, \dots, T$,

$$\sum_{m=1}^4 \hat{p}_m \hat{\phi}_y(\mathbf{x}_{it,y}, \hat{v}_{my}) > 0 \quad \text{and} \quad \sum_{m=1}^4 \hat{p}_m \hat{\phi}_l(\mathbf{x}_{it,l}, \hat{v}_l) > 0.$$

We find that the monotonicity and concavity conditions are satisfied for every $i = 1, \dots, N$ and $t = 1, \dots, T$.

Table 14 reports simulated labor supply elasticities both at the intensive and the extensive margins for the whole sample, but also by selected characteristics. The simulation algorithm with regard to labor supply elasticities is described in Appendix A.4. Panel a) of Table 14 focuses on the results from the benchmark model, i.e., the one including the unobserved heterogeneity $\mathbf{v}$. Panel b) reports instead labor supply elasticities after the estimation of the model without unobserved heterogeneity. Panel a) shows that intensive elasticities, which measure the percent change in working hours when the net income increases by 1%, are large. Across the whole sample, a 1% increase in net income implies an overall increase in working hours of 0.53%, corresponding to an increase of 6.65 yearly working hours on average, and an increase in the employment rate by 0.19% (0.13% points on average). Similar to the marginal rates of substitution of income with leisure, the labor supply elasticities are quite homogeneous across different demographic groups. The labor supply seems to be more rigid for women with tertiary education and for women without children, suggesting that they are more career oriented. When we estimate the model neglecting the unobserved component $\mathbf{v}$, we get a labor supply elasticity which is marginally smaller on average, as displayed in the first line of panel b). Also when the unobserved heterogeneity is neglected we get quite homogeneous labor supply elasticities across different demographic groups, apart from the labor supply elasticity of women in bad health.

In the evaluation of the effects of TRA86 on well-being, the estimated parameters of the utility function are used to simulate the choice of net income–leisure bundles of families under a counterfactual scenario in order to isolate the effect of TRA86. As the reliability of the simulations depends on the capability of our discrete choice structural labor supply model to predict the realized household choices between net income and leisure, we first report goodness-of-fit checks of the estimated model. The goodness-of-fit statistics are constructed on the basis of simulations of 9999 bundle choices for each family in the sample and for each year the family remains in the sample. Since we replicate the simulations 9999 times, we can construct 95% confidence intervals of the predicted frequencies of working hours and check whether the predicted frequencies are close enough to the empirical ones. The simulation algorithm with regard the goodness of fit is in “Appendix A.4”.

The first panel of Table 15 contrasts the empirical frequencies of the yearly working hours discrete categories with the simulated ones across the whole time window 1984–1990. The bottom panel of Table 15 reports the model fit for two particular years, the first one (1984) and the last one (1990) of our time window. The aim is to check whether the estimated model is able to replicate the actual data over time although time enters the model specification in a very limited way (only through the wage equation and

Table 14 Simulated labor supply elasticities

Characteristics	Yearly variation in total hours	Intensive elasticity (%) ^a	Variation in employment rate (percentage points) ^b	Extensive elasticity (%) ^c	# of pooled observations
<i>(a) With unobserved heterogeneity v</i>					
Whole sample	6.649	0.526	0.129	0.188	13,275
Children between 0 and 6	6.644	0.565	0.158	0.218	5166
Children between 6 and 17	6.565	0.527	0.126	0.186	4765
No children	6.163	0.470	0.104	0.148	3344
Wife in bad health	6.674	0.555	0.140	0.208	1187
Wife in good health	6.477	0.523	0.128	0.186	12,088
Wife younger than 35	6.581	0.542	0.140	0.206	7346
Wife older than 35	6.388	0.507	0.116	0.167	5929
With tertiary education	5.621	0.421	0.082	0.112	5515
Without tertiary education	7.116	0.612	0.162	0.245	7760
White	6.530	0.530	0.130	0.190	10,015
Non-white	6.385	0.514	0.125	0.182	3260
<i>(b) Without unobserved heterogeneity v</i>					
Whole sample	5.961	0.479	0.142	0.161	13,275
Children between 0 and 6	6.286	0.623	0.198	0.240	5166
Children between 6 and 17	5.965	0.447	0.121	0.134	4765

Table 14 continued

Characteristics	Yearly variation in total hours	Intensive elasticity (%) ^a	Variation in employment rate (percentage points) ^b	Extensive elasticity (%) ^c	# of pooled observations
No children	5.453	0.368	0.084	0.092	3,344
Wife in bad health	2.021	0.234	0.064	0.086	1,187
Wife in good health	6.348	0.495	0.149	0.168	12,088
Wife younger than 35	6.007	0.490	0.151	0.173	7,346
Wife older than 35	5.905	0.466	0.130	0.147	5,929
With tertiary education	6.998	0.521	0.156	0.171	5,515
Without tertiary education	5.225	0.445	0.132	0.154	7,760
White	5.750	0.481	0.148	0.171	10,015
Non-white	6.610	0.473	0.122	0.133	3,260

^a Labor supply intensive elasticities measures the percentage increase in labor supply when net income increases by 1%

^b The variation in employment rate measures the change in the employment rate in percentage points when net income increases by 1%

^c Labor supply extensive elasticities is the percentage increase in employment rates when net income increases by 1%

Table 15 Goodness of fit: empirical and predicted frequencies, 1984–1990

Yearly working hours categories (h)	Actual frequencies ^a	Predicted frequencies ^b	95% confidence interval ^c	
<i>Pooled from 1984 to 1990</i>				
0	0.207	0.276	0.192	0.427
(0, 750]	0.130	0.041	0.015	0.080
(750, 1250]	0.108	0.077	0.033	0.117
(1250, 1750]	0.154	0.180	0.094	0.224
(1750, 2250]	0.328	0.338	0.224	0.495
(2250, 4000]	0.073	0.088	0.059	0.125
<i>1984</i>				
0	0.215	0.293	0.204	0.468
(0, 750]	0.147	0.050	0.020	0.090
(750, 1250]	0.118	0.094	0.043	0.136
(1250, 1750]	0.161	0.194	0.099	0.249
(1750, 2250]	0.292	0.297	0.211	0.457
(2250, 4000]	0.067	0.072	0.046	0.104
<i>1990</i>				
0	0.193	0.283	0.200	0.450
(0, 750]	0.121	0.044	0.014	0.091
(750, 1250]	0.102	0.085	0.033	0.134
(1250, 1750]	0.151	0.194	0.096	0.251
(1750, 2250]	0.342	0.331	0.225	0.500
(2250, 4000]	0.091	0.081	0.052	0.117

^a Empirical frequencies lying in the 95% confidence intervals of the simulated frequencies are in bold

^b The predicted frequencies are calculated using the sample characteristics and the ML estimates of the parameters of the labor supply model presented in Table 13

^c The 95% confidence intervals are computed from the predicted frequencies by sampling 9,999 times the vector of parameter estimates assuming that the estimator is normally distributed around the point estimates with a variance-covariance matrix equal to the estimated one

time-varying regressors).³⁰ The goodness of fit of the frequencies in 1984 and 1990 is in line with the model fit on the full sample.

A.4 Simulation algorithms with regard to marginal rates of substitution, labor supply elasticities and the goodness of fit

We start by describing the simulation algorithm for the goodness of fit, since it is the starting point of the algorithms for the estimation of the marginal rates of substitution and labor supply elasticities.

³⁰ We also tried to include time dummies in the linear preference terms of leisure and net income. Since their coefficients were not jointly significant at the usual 5% statistical level and the goodness of fit did not show any relevant improvement, we removed the time dummies from the set of covariates to save in degrees in freedom.

Simulation algorithm with regard to the goodness of fit

The simulation algorithm with regard to the goodness of fit proceeds according to the following steps:

1. Draw a vector of parameter estimates $\hat{\Theta}$ assuming normality around the point estimates with a variance-covariance matrix equal to the estimated one. This ensures that the Monte Carlo confidence intervals encompass the parameter estimation precision.
2. Assign to each family the observed explanatory variables, observed family net income for each wife's leisure choice and a vector of unobserved characteristics drawn with probabilities given by Eq. (8).³¹
3. For each bundle $j = 1, \dots, 6$ and each $i, \dots, N$, compute the predicted utility as the sum of the deterministic part of the utility function and a random draw ε_{ij} from the type I extreme value distribution, i.e., $U_{ij} = u(y_{ij}, l_j | \mathbf{x}_i, \hat{\mathbf{v}}_i; \hat{\Theta}) + \varepsilon_{ij}$.
4. Family i is predicted to choose the bundle j of net income and leisure when $U_{ij} > U_{ik}, \forall k \neq j$.
5. Repeat steps 1 to 4 $R = 9,999$ times to get R independent realizations and build Monte Carlo confidence intervals.

Simulation algorithm with regard to the marginal rates of substitution

The simulation algorithm with regard to the marginal rates of substitution is very similar to the one for the goodness of fit. The difference is that after step 4, i.e., after simulating the bundle choice of family i , we computed the variation in the level of income needed to reach the same level of predicted optimal utility if leisure would increase by 50 h/year (decrease of 50 h of work). Finally, we averaged across the sample to get the average marginal rate of substitution and we repeated the procedure 999 times to build Monte Carlo confidence intervals.

The average partial effects in Table 2 of selected characteristics on the marginal rate of substitution are computed by taking the variation in the simulated marginal rate of substitution when we let one particular covariate vary. If the covariate is a dummy variable, we measure the variation in the marginal rate of substitution when the covariate changes from 0 to 1. When the covariate has a continuous support (wife's age, husband's age, or the number of children), we look at the variation when it increases by the amount indicated in Table 2.

Simulation algorithm with regard to the labor supply elasticity

The labor supply elasticity predicted by the model is computed by looking at the change in the predicted discrete working hour points generated by a 1% increase in the net income corresponding to each discrete working hour point. We modified the simulation algorithm for the goodness of fit and replaced steps 4 and 5 with the following steps:

³¹ Since the unobserved heterogeneity $\mathbf{v}_i$ is a random term at household level, at each replication of the simulation algorithm, each individual is assigned a new vector $(\hat{v}_{yi}, \hat{v}_{li})$ of unobserved characteristics.

- 4' For each bundle $j = 1, \dots, 6$ and each $i, \dots, N$, compute also the counterfactual predicted utility with a 1% increase in net income $\tilde{U}_{ij} = u(1.01 \times y_{ij}, l_j | \mathbf{x}_i, \hat{\mathbf{v}}_i; \hat{\Theta}) + \varepsilon_{ij}$.
- 5' Family i is predicted to choose the bundle j of net income and leisure when $U_{ij} > U_{ik}, \forall k \neq j$. In the counterfactual scenario in which the disposable income is 1% larger, family i is predicted to choose the bundle j of net income and leisure when $\tilde{U}_{ij} > \tilde{U}_{ik}, \forall k \neq j$. On the basis of the predicted working hours without and with the 1% increase in net income, we calculate different measures of labor supply elasticity both at the extensive and the intensive margins by averaging across the sample (or across subgroups conditional on selected observed characteristics).
- 6' Repeat steps 1, 2, 3, 4' and 5' 999 times to get 999 independent realizations of the statistics of interest and build Monte Carlo confidence intervals.

A.5 Simulation algorithms to calculate welfare metrics

In order to understand the effect of TRA86 on households' choices and, thereby, on the attained well-being level and on the position in the well-being ranking, we predicted twice the choices of the households in our sample in 1986 using the first 4 steps of the simulation algorithm used for the goodness of fit in Sect. A.4: first, using the actual 1986 tax rule to go from gross to net incomes, getting the optimal bundle (l^*, y^*) ; secondly, using the 1988 tax rule to transform 1986 gross incomes into net incomes, getting the optimal bundle (l', y') . By substituting these bundles into the estimated utility functions, we get the indirect utilities attained in the actual and counterfactual scenarios. Since the welfare metrics used in our empirical analysis are invariant to an additive rescale of the utility function, it does not matter whether we compute them on the basis of the indirect utility $U(y^*, l^*; \mathbf{x}, \mathbf{v}, \varepsilon) = u(y^*, l^*; \mathbf{x}, \mathbf{v}) + \varepsilon_l$ or on the basis of its deterministic component $u(y^*, l^*; \mathbf{x}, \mathbf{v})$.

Calculation of the money-metric utility with $\tilde{w} = 0$

Once we get the indirect utilities from the first three steps of the simulation algorithm for the goodness of fit, the computation of the money-metric utility with $\tilde{w} = 0$ for individual i boils down to the calculation of the vertical intercept in the space (h, y) of the indifference curve reached by individual i (see Fig. 3), i.e.,

$$m_i^*(\tilde{w} = 0, y_i^*, l_i^*) = \left[1 + \frac{\hat{\alpha}_l}{\hat{\alpha}_y} \hat{u}(y^*, l^*) \right]^{1/\hat{\alpha}_l} \quad \text{for the actual tax scenario;} \quad (\text{A.4})$$

$$m_i'(\tilde{w} = 0, y_i', l_i') = \left[1 + \frac{\hat{\alpha}_l}{\hat{\alpha}_y} \hat{u}(y', l') \right]^{1/\hat{\alpha}_l} \quad \text{for the counterfactual tax scenario.} \quad (\text{A.5})$$

Calculation of the money-metric utility with $\tilde{w} > 0$

To compute the money-metric utility with $\tilde{w} > 0$, we must first identify the bundle (l_0, y_0) located along the indifference curve attained by individual i with optimal choice (l^*, y^*) (or (l', y') in the counterfactual scenario), where the indifference curve has slope equal to $\tilde{w}$. In other words, we have to solve a system of two equations for y_0 and l_0 : the marginal rate of substitution between l and y equal to $\tilde{w}$ and the indirect utility at the optimal bundle equal to the usual formula in Eq. (3) for (l_0, y_0) :

$$\begin{cases} \frac{\hat{\phi}_l}{\hat{\phi}_y} \frac{l_0^{\hat{\alpha}_l - 1}}{y_0^{\hat{\alpha}_y - 1}} = \tilde{w} \\ \hat{u}(y^*, l^*) = \hat{\phi}_y \frac{y_0^{\hat{\alpha}_y} - 1}{\hat{\alpha}_y} + \hat{\phi}_l \frac{l_0^{\hat{\alpha}_l} - 1}{\hat{\alpha}_l} \end{cases} \quad (\text{A.6})$$

$$\hat{u}(y^*, l^*) = \hat{\phi}_y \frac{y_0^{\hat{\alpha}_y} - 1}{\hat{\alpha}_y} + \hat{\phi}_l \frac{l_0^{\hat{\alpha}_l} - 1}{\hat{\alpha}_l} \quad (\text{A.7})$$

Solving Eq. (A.6) for y_0 and substituting into Eq. (A.7) yield

$$\frac{\hat{\phi}_y}{\hat{\alpha}_y} \left[\left(\frac{\hat{\phi}_l}{\hat{\phi}_y} \frac{l_0^{\hat{\alpha}_l - 1}}{\tilde{w}} \right)^{\frac{\hat{\alpha}_y}{\hat{\alpha}_y - 1}} - 1 \right] + \hat{\phi}_l \frac{l_0^{\hat{\alpha}_l} - 1}{\hat{\alpha}_l} - \hat{u}(y^*, l^*) = 0, \quad (\text{A.8})$$

which cannot be solved analytically for l_0 . Hence, we numerically minimize the absolute value of Eq. (A.8) with respect to l_0 and, by substituting the solution into Eq. (A.6), we get the solution for y_0 . If $l_0 < 0$ or $l_0 > 1$, we are in the presence of a corner solution and l_0 is replaced by $l_0 = 0$ or $l_0 = 1$, respectively.

Once we know the location of (l_0, y_0) , the money-metric utility with $\tilde{w} > 0$ is the vertical intercept of the budget set passing through (l_0, y_0) with slope $\tilde{w}$. Hence, $m_i^*(\tilde{w}, y_i^*, l_i^*) = y_0 - \tilde{w}(1 - l_0)$.

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